TINMANS TUNING GUIDE



FFF / FDM FILAMENT CALIBRATION

Tinmans Tuning Guide

A printer-neutral handbook for filament calibration, diagnosis, profile release, and practical OrcaSlicer/TinmanX1 settings.

Project initiated and experimentally informed by William Tinney. Researched, written, organized, and published with Codex by OpenAI.

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Tinmans Tuning Guide

A source-backed, printer-neutral handbook for building reliable FFF/FDM filament profiles.

Download the complete linked PDF

Slicer downloads:

- Download the latest official OrcaSlicer release
- Download the latest public TinmanX1 release

Release pages move forward over time. The settings reference states the exact versions and source revisions used for this edition.

This guide explains what to tune, why the order matters, how to read each test, and how to preserve evidence so a good profile can be reproduced instead of rediscovered. It is written for ordinary Cartesian, CoreXY, delta, bedslinger, direct-drive, Bowden, single-tool, and multi-tool printers. Slicer and firmware names vary, but the physical problems are the same.

Start Here

If the printer is mechanically sound and the filament is dry, use this order:

1. Define the baseline.
2. Verify profile, tool, and nozzle identity.
3. Condition the filament and tune temperature.
4. Find a safe maximum volumetric flow.
5. Tune pressure or linear advance.
6. Tune global and feature-specific flow.
7. Balance cooling, bridges, and retraction.
8. Correct dimensions and fit.
9. Refine seams and surfaces.
10. Separate acceleration, input-shaping, and VFA evidence.
11. Validate and release the profile.

For a symptom-first route, use the diagnostic matrix. For unfamiliar terms, use the glossary. For plain-language slicer controls, start with the OrcaSlicer and TinmanX1 settings reference.

The Short Version

1	Machine baseline
2	Dry and identify filament
3	Temperature
4	Maximum volumetric flow
5	Pressure or linear advance
6	Global flow ratio
7	Cooling and bridges
8	Retraction and travel
9	Dimensions and tolerances
10	Seams and surfaces
11	Representative validation print

Later tests depend on earlier ones. A temperature change can alter melt flow, pressure response, stringing, and layer adhesion. A pressure-advance change can alter corners and seams. Dimensional compensation should therefore come after the extrusion system is stable.

Five Rules That Prevent Most Tuning Mistakes

1. Verify the physical tool, nozzle, resolved profiles, and final G-code agree.
2. Change one causal variable at a time after any broad search.
3. Keep a control sample and record the complete conditions.
4. Tune extrusion quality before dimensional compensation.
5. Use the setting that matches the error model.

That fifth rule is crucial:

- Use flow ratio to correct how much material is deposited.
- Use pressure advance to correct pressure lag during speed changes.
- Use shrinkage compensation for proportional cooled-part scale error.
- Use contour or hole compensation for near-constant offsets.
- Use elephant-foot compensation only for enlarged bottom layers.
- Do not alter axis calibration to hide filament shrinkage.

What This Guide Does Not Promise

There is no universal best temperature, flow ratio, retraction distance, or pressure-advance value. The result depends on the exact filament, color, lot, moisture state, nozzle, hotend, extruder, layer geometry, speed, acceleration, cooling, and chamber conditions. This guide provides a method for finding and defending a value for a defined setup.

Safety

- Treat hotends, beds, chambers, and freshly printed parts as burn hazards.
- Keep hands, tools, hair, and clothing clear of moving machinery.

- Follow the filament manufacturer's safety data and ventilation guidance.
- Do not leave an unproven material or profile unattended.
- Some materials release irritating or hazardous emissions. Use suitable local exhaust or enclosure ventilation for the material and environment.
- Dry filament only within the limits of its spool, packaging, and manufacturer.

Repository Contents

File	Purpose
How to use this guide	Test design, evidence, and decision rules
Foundations	Mechanical, slicer, first-layer, and bed-adhesion baseline
Conditioning and temperature	Moisture, temperature towers, adhesion
Volumetric flow	Melt-capacity ceiling and speed conversion
Pressure advance	Pressure lag, test reading, firmware notes
Flow ratio	Global and feature-specific extrusion
Cooling and retraction	Cooling tradeoffs, bridges, travel artifacts
Dimensional accuracy	Scale, offsets, holes, elephant foot
Seams and surfaces	Seam diagnosis and surface refinement
Validation	Acceptance tests and profile release
Material families	Material-specific starting priorities
Settings reference and versions	Scope, version basis, and Orca/TinmanX1 difference labels
Quality tab	Geometry, walls, surfaces, seams, and precision
Strength tab	Shells, infill, structural paths, and Strength Lens
Speed tab	Feature speeds, acceleration, jerk, and transitions
Support tab	Conventional, tree, raft, and TinmanX1 Wave Overhang support
Multimaterial tab	Prime towers, assignments, purge, and interlocking
Others tab	Adhesion, special modes, output, and scripts
TinmanX1 fiber and strength tools	Process, filament, and machine fiber controls
Motion diagnostics	Acceleration towers, input shaping, VFA, and a worked field example
Profile integrity and experimental resets	Tool/nozzle identity, false baselines, machine-specific profiles, and a photographed FibreSeek reset
Diagnostic matrix	Symptom-to-cause triage
Glossary	Definitions and formulas
Experiment log	Reusable test record
Release checklist	Final profile audit
Sources	Primary references and further reading
Credits	Attribution and authorship
Complete PDF	Single-file publication with linked contents and cross-references

Evidence Standard

Every accepted value should have:

- a dated experiment record;
- the exact printer, tool, nozzle, material, and filament condition;
- the exact project or sliced G-code;
- a list of fixed variables and the one changed variable;
- photographs in a repeatable orientation;
- measurements taken after the part fully cooled;
- an explicit decision: `candidate`, `accepted`, `rejected`, or `superseded`.

The experiment template is designed for this.

Rebuild the PDF

The PDF is generated from the repository's Markdown sources so the web and single-file editions stay synchronized.

```
python3 -m pip install -r requirements-pdf.txt
python3 scripts/build_pdf.py
```

The finished publication is written to `output/pdf/Tinmans-Tuning-Guide.pdf`.

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Project initiated and experimentally informed by William Tinney. Researched, written, organized, and published with Codex by OpenAI. No source author, project, or company listed here has endorsed this guide.

How to Use This Guide

Filament tuning is controlled experimentation. The goal is not to make one calibration model look perfect. The goal is to produce a profile that behaves predictably on real parts and can be explained, repeated, and maintained.

The Three Layers of a Profile

Keep these categories separate:

Category	Examples	Store the result in
Machine	steps or rotation distance, bed mesh, tool offsets, input shaping	printer or firmware profile
Filament	temperature, flow ratio, PA, cooling, MVS, shrinkage	filament/material profile
Process	layer height, line width, walls, infill, support, speed strategy	process/print profile

A symptom can cross categories, but a correction should live where its cause lives. A single spool should not redefine the machine's X-axis calibration.

Values in the filament layer are still machine-specific unless independently validated elsewhere. PA, MVS, retraction, cooling response, and temperature under flow can change with the extruder, hotend, nozzle, and enclosure. Use a separate compatibility-gated filament profile for each proven machine/tool/nozzle combination.

Prove the Test Identity

One-variable testing is meaningful only after the baseline identity is true. The physical tool and nozzle, printer profile, filament profile, process geometry, and final G-code must agree. A familiar preset name or settings-panel screenshot proves intent, not execution.

Run the profile-integrity gate before the first calibration and after nozzle, hotend, firmware, slicer, or profile changes.

Define the Question Before Printing

A useful test begins with one sentence:

> With every listed condition fixed, does changing **X** from **A** to **B** > improve **Y**, measured by **Z**?

Example:

> With temperature, global flow, pressure advance, speed, and geometry fixed, > does increasing top-surface flow from 1.00 to 1.04 close the visible channels > without creating ridges or wall buildup?

If the test cannot be stated this way, it probably changes too many variables.

Broad Search, Then Fine Search

Use a broad test to find the useful region, then narrow the increments:

1. Search a safe range with large steps.
2. Reject clearly bad regions.
3. Repeat around the best region with smaller steps.
4. Confirm the selected value on a separate print.

This is faster and more reliable than making many tiny changes from an unknown starting point.

Keep a Control

Preserve either the previous accepted specimen or an unchanged control in the same print. A control reveals whether room temperature, filament moisture, hardware state, or a slicer update changed between sessions.

For visual tests:

- use the same material color and lighting;
- photograph the same faces in the same orientation;
- label samples physically;
- avoid judging glossy material from a single flash angle.

Freeze the Test Conditions

Unless the selected variable requires a change, hold these constant:

- printer, tool, nozzle, and nozzle wear state;
- build plate, cleaning procedure, and adhesive;
- filament spool, drying cycle, and feed path;
- chamber, door, lid, ambient temperature, and draft exposure;
- layer height, line width, wall count, infill, and model orientation;
- temperature, speed, acceleration, cooling, and retraction;
- bed mesh, Z offset, and tool offsets;
- slicer version and profile inheritance.

Record Evidence, Not Memory

Save the complete project and final G-code when possible. A screenshot of a settings panel is not enough because inherited or hidden settings can alter the slice.

Use four result states:

- `candidate`: promising but not independently confirmed;
- `accepted`: repeatable and promoted to the profile;
- `rejected`: tested and unsuitable under the recorded conditions;
- `superseded`: once accepted, later replaced with evidence.

The experiment log captures the minimum useful record.

If a later audit finds that the wrong nozzle, tool, or resolved profile was used, preserve the observation but change its status to `rejected - invalid baseline`. Such a trial can show how a control influences a defect, but its numerical value cannot be promoted to the intended profile.

Measure Carefully

- Let the part fully cool before measuring.
- Use a suitable instrument and confirm its zero.
- Measure several locations, not the seam, corners, or elephant foot.
- Repeat the measurement and report resolution honestly.
- Use a larger artifact when a small dimensional error approaches the tool's uncertainty.
- Test both external and internal features before claiming a tolerance.

A `0.02 mm` difference measured once with ordinary calipers is not necessarily real. A repeatable `0.13 mm` difference on multiple axes deserves attention.

Separate Observation from Interpretation

Write both:

> Observation: parallel open channels remain between top lines.
> Interpretation: top-surface deposition may be low; global under-extrusion is less likely because the walls are continuous and dimensions are correct.

This preserves useful evidence even when the first interpretation is wrong.

Know When to Stop

Stop tuning when the profile meets the defined requirements with margin. A functional enclosure and a display model can need different profiles for the same spool. Chasing an invisible difference can reduce strength, speed, or reliability.

Foundations: Establish a Trustworthy Baseline

Filament calibration cannot compensate for a loose printer, a partially blocked nozzle, an unstable bed, or an incorrect tool offset. Prove the machine first.

Hardware Checklist

Inspect and record:

- nozzle diameter, material, installation, contamination, and wear;
- hotend assembly, heater, temperature sensor, and fan operation;
- extruder gear cleanliness, tension, alignment, and filament path drag;
- belts, pulleys, fasteners, rails, wheels, bearings, and lubrication;
- build plate type, flatness, attachment, cleanliness, and damage;
- part-cooling duct position and fan response;
- chamber sensors, doors, lids, and exhaust behavior;
- spool path, reverse Bowden tube, buffer, or material changer resistance.

Filled or abrasive materials can enlarge a soft nozzle. A worn nozzle changes line geometry and may make a formerly valid profile appear over-extruded.

Firmware and Motion Checklist

Confirm:

- X, Y, Z, and extruder motion calibration are mechanically correct;
- homing and endstop behavior are repeatable;
- bed tramming and mesh probing are valid at the intended temperature;
- the expected mesh and tool offsets are actually loaded;
- input shaping or resonance compensation is appropriate for the machine;
- acceleration and speed limits do not cause skipped steps;
- thermal protection is enabled and temperature control is stable.

Extruder rotation distance or E-steps is a machine relationship between motor motion and commanded filament feed. Establish it with the extruder operating in the manufacturer's supported manner. Do not retune it for every spool.

First Layer

The first layer must be repeatable before material tuning:

- adjacent lines touch without deep ridges;
- the line adheres through corners and direction changes;
- the nozzle does not scrape the plate;
- the bottom is not so compressed that it creates a large elephant foot;
- the plate reaches a stable temperature before probing and printing.

Do not judge global flow from a first layer. First-layer height, Z offset, bed texture, and initial-layer flow deliberately change its appearance.

Build-Plate Preparation and Adhesion

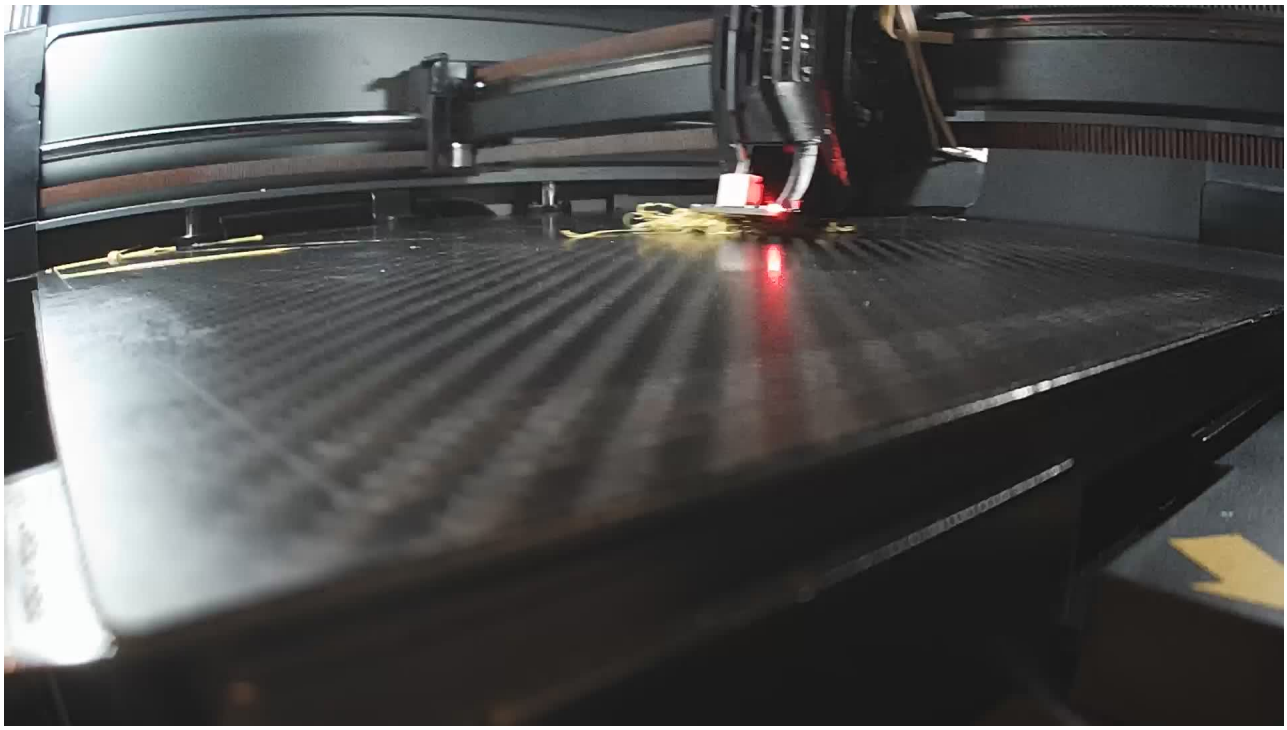
Treat plate preparation as a controlled tuning input. A calibration tower that releases from the plate is not valid evidence for temperature, maximum flow, pressure advance, or flow ratio above the release point.

Record the exact plate, cleaning method, adhesive and application state, bed temperature, first-layer controls, and brim with every calibration. If the part moves, stop and correct adhesion before changing unrelated filament settings.

Tinman field result: R3Men carbon-fiber plate

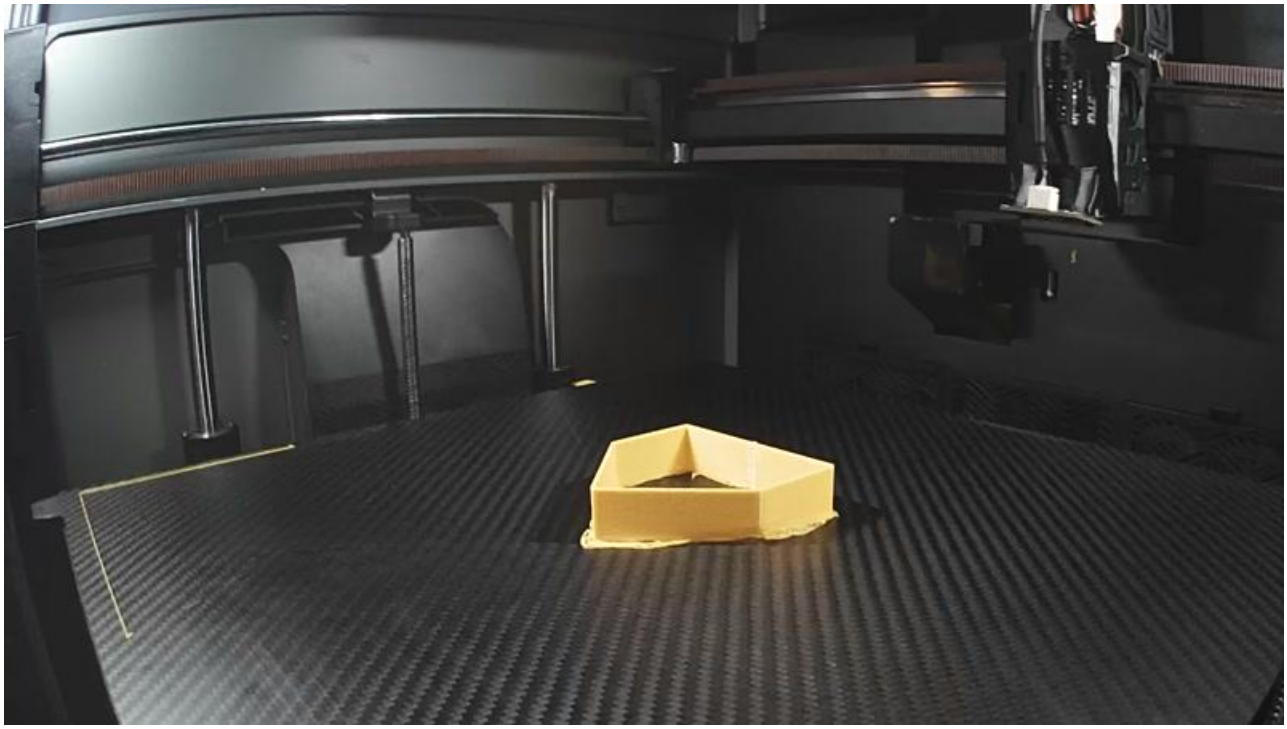
On the R3Men carbon-fiber build plate used in the Tinman fleet, Vision Miner Nano Polymer is the only bed adhesive we have found to work consistently. This is a field result for the tested plates, materials, and procedures, not a universal compatibility claim.

The photographed K2 Plus pressure-advance test below used dry 3D-Fuel PCTG, a `0.6 mm` nozzle, an `85 C` bed, and the same hash-verified G-code in every run. Three attempts released by PA `0.018`, including one after the Beacon cover was secured and the plate had been cleaned. The failed third attempt formed the filament nest shown here.



K2 Plus PA tower detached from an R3Men carbon-fiber plate at PA 0.018 during the third controlled attempt. The Beacon cover had been secured and the plate cleaned, but the small-footprint specimen released. Field camera frame by William Tinney, 2026.

For the fourth attempt, the operator thoroughly washed the plate with warm water and dish soap, rinsed and dried it, then applied a fresh coat of Vision Miner Nano Polymer. The printer, material, profile, `5 mm` outer-only zero-gap brim, and PA `0.016-0.048` schedule were unchanged. The tower remained attached through PA `0.048` and completed normally.



The unchanged K2 Plus PCTG PA tower completed and remained attached after a thorough warm soap-and-water wash and fresh Vision Miner Nano Polymer application. Field camera frame by William Tinney, 2026.

The lesson is not that adhesive can compensate for a dirty plate. Adhesive presence alone did not make the earlier attempts reliable. The repeatable procedure is:

1. Let the plate reach a safe handling temperature and remove all failed-print residue.
2. Wash thoroughly with warm water and residue-free dish soap, rinse well, and dry with a clean lint-free material. Do not touch the print area afterward.
3. Apply a fresh, even Nano Polymer coating according to its product instructions.
4. For a small-contact calibration object, use a `5 mm` outer-only, zero-gap brim and verify the emitted brim actually joins the model.
5. Keep the preparation identical for confirmation and production prints, and log any change as an experimental variable.

Camera evidence can prove retention and completion. It cannot by itself grade fine PA faces, surface finish, or layer strength; inspect the cooled part physically before promoting a profile value.

Temperature Stability and Heat Soak

High-temperature materials and large enclosed printers can change as the frame, bed, chamber, and toolhead warm. Use the same heat-soak procedure for calibration and production. Record either the soak time or the condition that starts the print, such as chamber temperature and bed stability.

Slicer Baseline

Create or duplicate a known working printer profile. Record:

- slicer name and exact version;
- firmware flavor and machine limits;
- tool and nozzle assignment;

- layer height and line widths;
- speed and acceleration by feature;
- wall generator and wall order;
- seam strategy;
- top/bottom layers and surface pattern;
- cooling and minimum-layer-time behavior;
- any start G-code that changes mesh, offsets, PA, flow, or temperature.

Calibration generators can leave temporary overrides in a project. Start a fresh project after using them and verify the resulting G-code.

Nozzle and profile identity gate

Do not infer nozzle diameter from the profile name. Verify the installed tool, the printer definition, process line widths, and emitted G-code as separate pieces of evidence. If they disagree, stop tuning and follow Profile Integrity and Experimental Resets.

A wrong nozzle assumption can make PA, seam, retraction, line-width, and flow experiments appear internally consistent while every accepted number belongs to a setup that does not physically exist.

Filament Identity

Give each profile a precise identity:

> Manufacturer / product / material / color / diameter / lot / date opened

Colorants and additives can change flow, temperature, and cooling behavior. Treat a different color or lot as unverified until a short confirmation print passes.

Baseline Acceptance Print

Before advanced tuning, print a simple artifact that includes:

- a broad wall;
- a top surface;
- a short bridge;
- an overhang;
- a seam or repeated corner;
- a known external dimension.

The purpose is not perfection. It confirms that the printer can complete a controlled test and reveals obvious mechanical or thermal faults.

If probing, homing, nozzle condition, or tool selection is not repeatable, log that as a machine event and suspend filament tuning. Do not count failed starts or hardware faults as material outcomes.

Filament Conditioning and Temperature

Temperature tuning is meaningful only when the material condition is stable. Water in filament can create defects that resemble poor temperature, flow, retraction, or pressure advance.

Moisture Symptoms

Possible signs include:

- popping, crackling, or visible vapor at the nozzle;
- rough, foamy, or inconsistent extrusion;
- bubbles, pits, blobs, or smoke-like wisps;
- excessive stringing despite reasonable retraction;
- weak or brittle layers;
- an unstable surface finish from layer to layer.

These signs are not individually conclusive, but drying is the correct first experiment when storage history is unknown.

Drying Safely

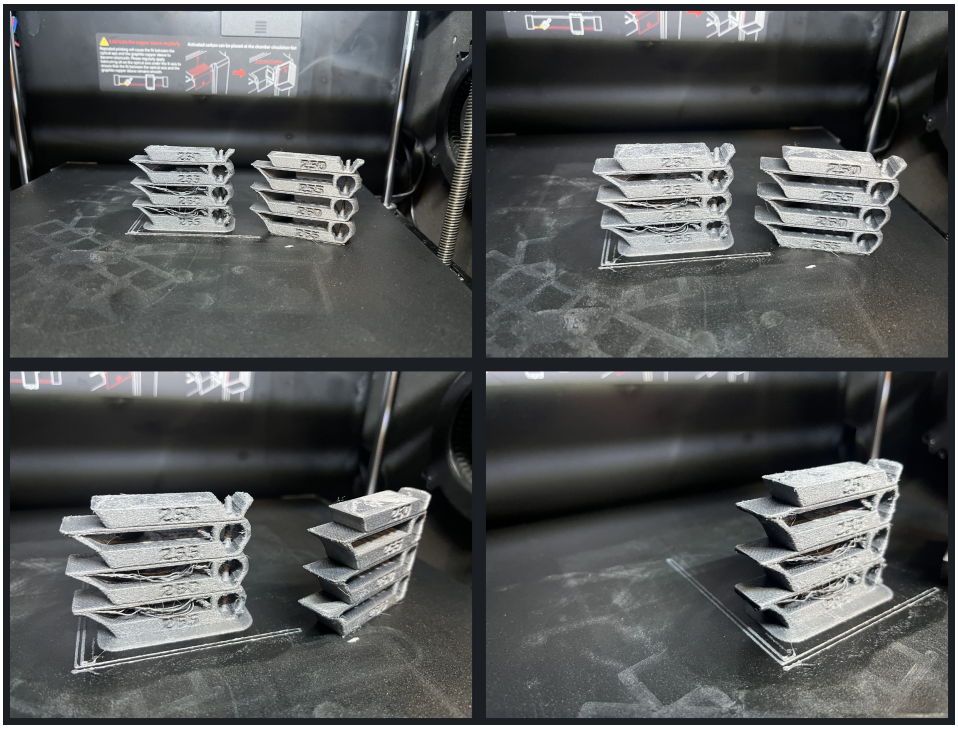
Use the filament manufacturer's recommendation when available. Confirm that the spool and any label, bag, RFID tag, or hub can tolerate the dryer temperature. Prusa's drying table is a useful cross-check, not permission to exceed a product limit.

Record:

- dryer type and measured temperature;
- drying duration;
- spool starting condition;
- time between drying and printing;
- whether the spool remained in a dry box during the test.

Controlled PET-CF Drying Comparison

The photographs below compare Elegoo PET-CF printed from the same G-code on the same Qidi Plus 4, nozzle, settings, and room conditions. The operator reported the left tower as undried and the right tower as dried. The dried specimen has markedly cleaner bridges, fewer loose strands, more coherent overhangs, and a more uniform surface. This is a controlled A/B result for one spool and process, not a universal drying schedule.



Elegoo PET-CF temperature-tower comparison. In every paired view, the undried specimen is on the left and the dried specimen is on the right. Same G-code, printer, nozzle, settings, and room conditions. Field photographs by William Tinney, 2026.

The comparison is also a useful diagnostic warning: moisture can produce defects that look like excessive temperature, poor retraction, weak bridging, or unstable flow. Dry and stabilize the spool before changing several slicer variables to chase those symptoms.

What Nozzle Temperature Changes

Higher temperature generally lowers melt viscosity and can improve high-flow extrusion and layer bonding. It can also increase ooze, stringing, gloss, degradation, and loss of small-feature definition.

Lower temperature may sharpen details and reduce ooze, but can cause poor bonding, incomplete melting, high backpressure, skipped extrusion, or a matte under-extruded surface.

The useful temperature therefore depends on flow rate and purpose. A slow display print and a fast structural print may need different values.

Design a Temperature Tower

1. Begin inside the manufacturer's safe range.
2. Use steps large enough to reveal a trend, commonly `5 C`.
3. Keep layer height, width, speed, cooling, and geometry fixed.
4. Verify in G-code preview or text that temperature commands occur at the intended heights.
5. Label each segment in the model or experiment record.
6. Include a range above and below the expected optimum when safe.

A tower does not identify an optimum if the best section is at the hottest or coldest tested endpoint. Extend the range in that direction.

Read the Tower

Evaluate every segment for:

- consistent walls and surface finish;
- overhang edge definition and curl;
- bridge sag and strand coherence;
- stringing and wisps;
- small-feature and text fidelity;
- corner quality and seams;
- discoloration or signs of degradation;
- layer adhesion after cooling.

Photographs are useful for appearance, but they cannot prove layer strength. Use a controlled bend or break comparison when adhesion matters. Keep geometry and loading consistent and use eye protection.

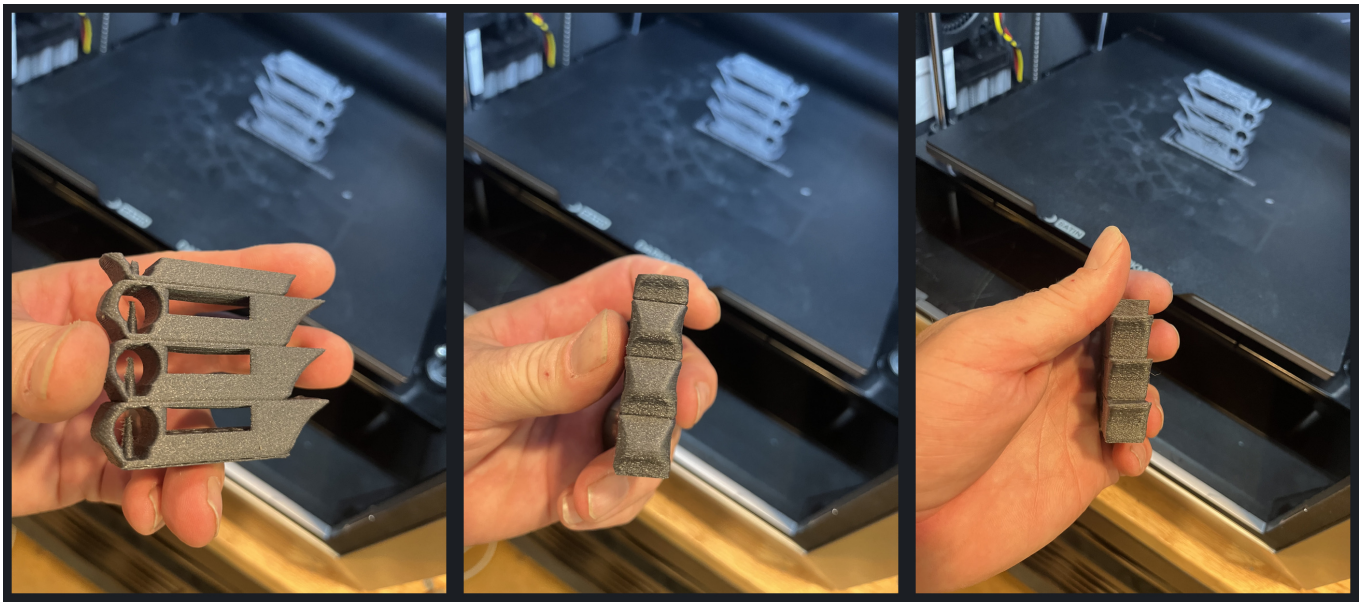
Selection Rule

Choose the lowest temperature that provides the required adhesion and stable flow at the intended production rate, unless a hotter setting produces a materially better mechanical result with acceptable detail and stringing.

For high-flow work, confirm the selected temperature again during the volumetric-flow test. A visually good low-speed tower section may not melt fast enough for production.

Preserve Margin Above a Cold-Flow Failure

The dried Elegoo PET-CF tower below ran from `265 C` at the bottom through `260 C`, `255 C`, and `250 C`. Two attempts stopped at approximately the same point as the program moved toward the next, colder step. After cancellation, the same filament path extruded `300 mm` normally after the nozzle was heated to `300 C`. That result is consistent with insufficient melt capacity at the low-temperature endpoint rather than a persistent clog.



Dried Elegoo PET-CF tower details after removal from the plate. The repeated tower stopped below the completed 250 C tier; the 255 C tier provides the best balance of surface quality and temperature margin for the next controlled tests. Field photographs by William Tinney, 2026.

For this machine, `0.6 mm` nozzle, and test flow, `255 C` is the defensible working selection. The `250 C` tier is visually competitive, but it is only one `5 C` step above the repeatable loss-of-flow boundary. The `255 C` tier retains clean walls, readable detail, and acceptable bridge behavior while preserving process margin. Treat that value as a test-specific starting point and confirm it again at the intended production volumetric flow.

First-Layer Temperature

A hotter first layer can improve wetting and reliability, but it is a separate setting from normal-layer temperature. Excess heat can increase ooze, elephant foot, or polymer degradation. Tune it only after Z offset and plate preparation are correct.

Common Misreads

Observation	Do not assume	Check
Stringing	temperature is too high	moisture, travel, retraction, nozzle residue
Matte upper tower	temperature is too low	whether flow rate also increases with height
Weak bridge	temperature alone is wrong	bridge fan, speed, flow, line width
Rough surface	over-extrusion	wet filament, partial clog, unstable temperature
Best result at endpoint	endpoint is optimum	extend the range safely

Once temperature is bracketed, find the system's usable volumetric-flow limit.

Maximum Volumetric Flow

Maximum volumetric flow is the amount of plastic the complete extrusion system can melt and deposit per second under defined conditions. It is not a universal hotend specification. Filament, temperature, nozzle geometry, extruder grip, and cooling all affect the result.

The Basic Equation

For a straight extruded path:

```
volumetric flow (mm^3/s) = speed (mm/s) * line width (mm) * layer height (mm)
```

Rearranged:

```
maximum speed = maximum volumetric flow / (line width * layer height)
```

Example:

```
MVS = 15 mm^3/s
line width = 0.45 mm
layer height = 0.20 mm
maximum steady extrusion speed = 15 / (0.45 * 0.20) = 166.7 mm/s
```

The machine may still be limited by acceleration, motion, minimum layer time, or feature-specific speed.

Why Tune MVS Before Flow Ratio

A flow-ratio test printed beyond the melt limit can look under-extruded even when the ratio is correct. Establish a safe throughput region first so later tests evaluate the requested setting rather than hotend saturation.

Printed Ramp Method

Use a tower or continuous artifact whose requested volumetric flow increases with height.

1. Use the accepted nozzle temperature and representative cooling.
2. Set a sufficiently high feature speed so MVS is the limiting factor.
3. Confirm the actual flow range in the slicer preview.
4. Print while watching for the first sustained change.
5. Measure the height of the onset and calculate the corresponding flow.

Failure evidence includes:

- matte or changed gloss that persists;
- thin, rough, or discontinuous lines;
- gaps between walls;
- missing material after acceleration;
- extruder clicking, skipping, or filament grinding;
- a sudden loss of layer adhesion;
- inability to hold nozzle temperature under load.

Ignore a single isolated blemish. Select the start of a sustained trend.

Gravimetric Extrusion Method

An optional bench method commands a known filament length at increasing rates, weighs the output, and compares actual mass with expected mass. This can reveal gradual flow drop-off before obvious print failure. It requires a reliable scale, known filament density or a consistent reference mass, safe collection of hot extrudate, and careful control of temperature and timing.

The printed-ramp method is simpler and includes motion and deposited-line behavior. The gravimetric method is useful for comparing hotends or diagnosing the extrusion system itself. Do not treat the two results as interchangeable.

Set a Production Limit

For a general-quality profile, set the slicer MVS below the first sustained quality loss. A practical starting margin is `5-10%`:

```
profile MVS = observed onset * 0.90 to 0.95
```

Use a larger margin for structural parts, long jobs, variable ambient conditions, or spools with inconsistent diameter. A speed-focused infill profile may accept more risk than an external-wall profile, but the tradeoff must be explicit.

Retest When These Change

- filament material, formulation, color, or moisture state;
- nozzle diameter, material, geometry, or wear;
- hotend, heater, heatbreak, or extruder;
- nozzle temperature;
- high-flow insert or nozzle;
- substantial layer-height or line-width regime;
- unusual backpressure from filled, flexible, or highly viscous material.

Strength Is a Separate Limit

The highest flow that looks continuous may not produce the strongest layer bond. If the part is structural, compare strength at the intended production flow and at a slower control. Surface appearance alone does not establish melt quality throughout the bead.

With a safe flow envelope established, tune pressure advance.

Pressure Advance and Linear Advance

Molten polymer, filament, Bowden tubes, and the extrusion path behave like a spring-damper system. When toolhead speed changes, nozzle pressure does not change instantly. Pressure advance compensates for that lag by adjusting extruder motion around acceleration and deceleration.

Common names include:

- Pressure Advance in Klipper and RepRapFirmware;
- Linear Advance or K-factor in Marlin;
- flow dynamics or similar vendor-specific terms.

The values are not portable between algorithms or firmware families.

What PA Can and Cannot Fix

Correct PA can reduce:

- bulging at corners and line ends;
- thin starts after acceleration;
- pressure-related seam blobs or gaps;
- inconsistent width during speed transitions.

PA does not correct:

- the total amount of material commanded;
- wet filament;
- a partial clog;
- backlash, loose belts, or resonance;
- a wrong dimensional scale;
- poor cooling or unsupported geometry.

Prerequisites

Before testing:

- settle temperature;
- stay below the safe MVS ceiling;
- use a mechanically sound extruder and nozzle;
- disable conflicting slicer pressure tricks where the firmware documentation requires it;
- choose representative speed and acceleration;
- confirm which firmware or slicer owns the active PA value.

Test Methods

Line or pattern test

Alternates slow and fast segments across a range of PA values. It is fast and material-efficient but depends heavily on first-layer quality.

Tower test

Changes PA with height and shows repeated corners. It is slower but easier to read when first-layer variation would obscure a line test.

Adaptive test

Some slicers can model PA across multiple flow and acceleration conditions. This is useful only when the hardware, firmware, and slicer support it and when a single value is demonstrably inadequate. Establish a good fixed PA first.

Read the Result

Appearance	Likely interpretation
Corner protrudes or remains swollen	PA too low
Line becomes thin before or after speed change	PA too high
Corner is sharp with continuous adjacent line	useful region
Only layer-change corner is damaged	seam/retraction contamination; read other corners
All values look poor	wrong range, flow saturation, mechanical problem, or wet filament

Klipper specifically advises ignoring the corner with the seam when reading its tower. When several values are equally good, choose the lower acceptable value; excessive PA can create subtle starvation and demands more extruder response.

Calculation Example

If a tower starts at `0.000`, increases by `0.002` per millimeter, and the best region is `8 mm` above the start:

```
PA = start + step * height
PA = 0.000 + 0.002 * 8 = 0.016
```

Use the exact generator formula. Some tests change per layer rather than per millimeter.

Dependence on Print Conditions

PA can change with:

- material and moisture;
- temperature and volumetric flow;
- nozzle and hotend;
- direct-drive versus Bowden path;
- acceleration and requested speed;
- flexible or highly compressible filament.

Retest after a meaningful upstream change. If a production profile spans a very wide flow and acceleration range, validate the compromise at slow external walls and fast internal features.

Do Not Judge PA by the Seam Alone

Seam placement, retraction, wipe, travel distance, ooze, and layer change can hide an otherwise correct PA value. Use non-seam corners for the initial PA decision, then return to seam tuning later.

Global and Feature-Specific Flow

Flow ratio, also called extrusion multiplier, scales the amount of material the slicer commands. Tune it for continuous, well-formed extrusion. Do not use it as a general dimensional correction.

Global Flow Versus Volumetric Flow

These are different controls:

- **Flow ratio** scales commanded material, such as `0.98` for `98%`.
- **Maximum volumetric flow** limits how quickly material is requested.

A correct flow ratio can still fail above the hotend's melt capacity. A low MVS cannot repair an incorrect flow ratio; it only keeps requests inside a safer rate.

Use a Broad Top Surface

A useful visual test has:

- enough bottom layers and infill to support the top;
- several top layers;
- a broad region with long, parallel top lines;
- known top line width and moderate speed;
- enough perimeter distance to reveal wall-interface behavior.

Inspect with both sight and touch under glancing light.

Too low

- open channels or dark valleys between lines;
- pinholes or incomplete closure;
- top lines separate from the perimeter;
- roughness caused by unsupported strands;
- weak, easily separated walls when other causes are excluded.

Useful region

- continuous lines with no broad open channels;
- an even tactile surface;
- clean contact with perimeter walls;
- no repeated nozzle plowing or material pileup.

Too high

- raised ridges that catch a fingernail;
- nozzle scars or plowing;
- material accumulating at line ends and walls;
- rough, overfilled corners;
- dimensional swelling across otherwise simple features.

Microscopic gaps visible only under extreme magnification can be normal. Judge whether the surface meets its functional purpose.

Coarse and Fine Passes

1. Start from a sane material profile.
2. Compare broad changes, often `1-2%`, to locate the useful region.
3. Refine around the best result in `0.5%` steps when needed.
4. Confirm on a fresh specimen.

If the starting ratio is `0.98` and a candidate modifier is `+2%`:

```
new ratio = 0.98 * 1.02 = 0.9996
```

Confirm the calibration generator's formula. Some tools add percentage points; others multiply the current value.

Measured Single-Wall Methods

Prusa documents a vase-wall measurement method. It can work with a suitable micrometer and controlled geometry, but ordinary calipers, rounded beads, corner pressure, wall flex, and slicer line-width models can introduce large relative error. Ellis' guide cautions against treating a single extruded wall as an exact rectangular bead.

Policy for this guide:

- prefer the visual/tactile top-surface method for routine flow tuning;
- use measured mass or carefully designed walls when the experiment calls for them and the uncertainty is understood;
- solve dimensions with dimensional tests, not a wall-flow shortcut.

Feature-Specific Flow

If walls and general extrusion are correct but one feature is deficient, use a feature-specific control where available:

- top or bottom solid infill flow;
- bridge flow;
- internal solid-infill flow;
- support-interface flow;
- first-layer flow.

Effective feature flow is often multiplicative:

```
effective top flow = global filament flow * top-surface flow ratio
```

Example:

```
global flow = 0.98
top flow = 1.04
effective top flow = 0.98 * 1.04 = 1.0192
```

This deposits `1.92%` more than nominal on top surfaces while leaving walls at the accepted global ratio.

First-Layer and Top-Surface Traps

A rough top can originate below the visible layer. Check:

- insufficient top-shell thickness;

- sparse or poorly supported infill;
- pillowing from heat and inadequate cooling;
- warped lower layers;
- first-layer over-compression carried through a thin object;
- an unsuitable top pattern, speed, or line width.

Do not raise global flow if only the top skin is deficient and the walls, dimensions, and other solid layers are correct.

Cooling, Bridges, Retraction, and Travel

Cooling and retraction both affect strings and small features, but they solve different physical problems. Tune cooling first for deposited geometry, then use retraction and travel controls for pressure and ooze during non-print moves.

Cooling Is a Tradeoff

More part cooling can improve:

- overhang definition;
- short bridges;
- small features and minimum-layer-time behavior;
- resistance to curl on newly deposited edges.

Too much cooling can cause:

- weak layer bonding;
- warping or cracking;
- inconsistent shrinkage;
- poor chamber stability;
- nozzle-temperature droop when airflow strikes the heater block.

Use the least cooling that delivers the required geometry and strength.

Separate Fan Roles

Printers may have several fans:

- hotend heatsink fan;
- part-cooling fan;
- auxiliary or side fan;
- chamber circulation fan;
- chamber exhaust fan.

Do not treat them as one percentage. An auxiliary fan can create a directional surface difference or warping that the main part fan does not.

Cooling Test Method

Use an artifact with repeatable overhang angles, small towers, and bridges. Hold temperature, flow, and speed constant.

Evaluate:

- overhang edge curl and underside shape;
- bridge sag, strand separation, and anchoring;
- small-feature deformation;
- layer adhesion after cooling;
- directional differences between faces;
- warping, lifting, or inter-layer cracks.

High-shrink materials should be tested in the same enclosure and chamber state used for production.

Bridge Controls

Bridges can have their own speed, flow, line width, cooling, PA, and direction. Tune them as a feature after general temperature and cooling are credible.

- Too much bridge flow can produce heavy sagging strands.
- Too little can produce thin, broken, or poorly anchored strands.
- Too fast can break anchoring or stretch strands excessively.
- Too slow can give the strand more time to sag.
- More fan can help shape but may hurt bonding or high-shrink materials.

Test the span lengths and directions that matter to the actual part.

Keep Bridge Evidence Separate from Normal Cooling

A tall temperature or flow tower may contain occasional bridging, but it does not replace a bridge-focused artifact. Use a compact bridge model with known span, line direction, and anchor geometry after temperature, MVS, PA, and flow are credible. Record the bridge-fan and bridge-speed overrides separately from the normal part-cooling value.

It is valid to use an aggressive bridge-only fan setting to find the geometry limit while keeping a lower normal fan value for layer bonding. Do not silently copy that stress-test fan value into a production profile. After choosing a bridge setting, confirm it on a representative part at the intended chamber condition.

Retraction Purpose

Retraction temporarily relieves or removes material from the nozzle entrance during travel. It cannot remove moisture, correct PA, or fix an excessively hot and leaking nozzle by itself.

Tune retraction only after:

- the filament is dry;
- temperature is accepted;
- PA is reasonable;
- travel moves are fast and mechanically reliable;
- the nozzle is clean and not partially clogged.

Retraction Test

1. Choose a range appropriate to the extruder path and manufacturer guidance.
2. Keep retraction speed fixed while tuning distance.
3. Select the shortest distance that gives the cleanest repeatable result.
4. Refine speed only if needed.
5. Confirm on a model with realistic travel distances and feature sizes.

Too much retraction can create:

- delayed restart or gaps after travel;
- filament grinding;
- heat-creep or clog risk;
- bubbles caused by drawing air into the melt zone;
- unnecessary wear and print time.

Travel, Wipe, and Z Hop

- **Travel speed** reduces ooze time but must remain mechanically reliable.
- **Avoid crossing walls/perimeters** can hide strings inside a part at the cost of longer travel.
- **Wipe** can reduce material left at a seam, but interacts with PA and retraction.
- **Z hop** can avoid collisions, but adds time and may increase ooze or surface marks. Use it for a demonstrated need.
- **Extra restart/prime** is rarely the first fix for a tuned modern extrusion system; it can trade a gap for a blob.

Diagnose Strings by Shape

Evidence	First suspects
Fine hairs everywhere	moisture, temperature, material tendency
Thick strings between towers	retraction distance/speed, travel time
Blob followed by string	seam pressure, PA, ooze, wipe
Gap after every travel	excessive retraction or insufficient restart
Strings only on one side	fan direction, airflow, travel path

Once deposited geometry and travel behavior are stable, move to Dimensional Accuracy.

Dimensional Accuracy and Tolerances

Dimensional tuning works only when the cause of the error is classified. A proportional scale error, a constant contour offset, an undersized hole, and an elephant foot require different controls.

Measure After Extrusion Is Stable

Settle temperature, MVS, PA, flow, and cooling first. Otherwise a geometry test can accidentally encode an extrusion defect into compensation.

Measurement protocol:

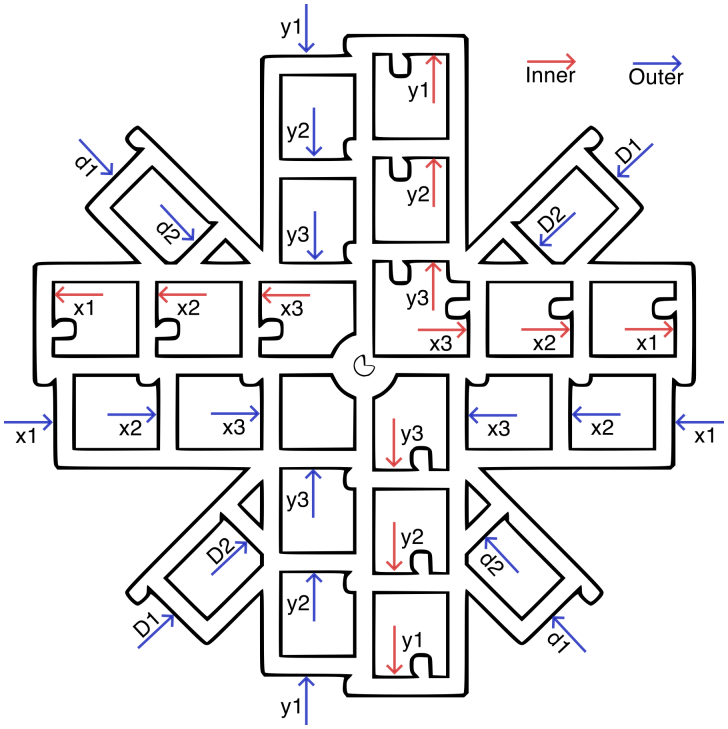
1. Let the part fully cool and condition consistently.
2. Zero and check the measurement tool.
3. Measure several points and both principal XY axes.
4. Avoid seams, corners, embossed text, and bottom flare.
5. Record raw values before averaging.
6. Use a larger test piece when the expected correction approaches the tool's uncertainty.
7. Include external and internal features.

Choose the Test by the Question

No one print can establish every kind of geometric accuracy. Use the artifact that separates the suspected causes instead of applying a correction from the most convenient print.

Question	Appropriate evidence	Do not conclude
Is the printer broadly producing a clean, recognizable part?	A Benchy or another geometry-rich benchmark; inspect hull, bridges, text, holes, overhangs, seams, and ringing.	That one Benchy dimension proves axis calibration or a global scale value.
Is cooled-part size proportional over several sizes?	A tiered dimensional stack with several nominal X/Y lengths; record the signed X and Y results for every tier.	That a single 20 mm cube is enough to distinguish flow, contour width, shrinkage, or motion.
Are X/Y scale, skew, and flow-related contour offset separable?	A purpose-built multi-point grid such as the open-source Calistar pattern; measure matched outer and inner spans at every marked location.	That a shared XY shrinkage value is valid when the axes disagree.
Does the machine move the commanded distance?	A dial indicator or other rigid external metrology while commanding raw axis motion.	That a printed plastic feature alone justifies changing rotation distance or steps.

Calistar 120 x 3 measurement layout



Calistar 120 x 3 measurement map. Blue arrows identify outer measurements; red arrows identify inner measurements. Record X1 through X3, Y1 through Y3, and both diagonal directions.

Figure attribution and license. This figure is rendered from the original, unmodified SVG measurement map from Calistar (formerly Fleur de Cali) by dirtdigger. It is distributed with the Calistar project under GPL-3.0; the source SVG and a copy of that license are included in this repository at `assets/third-party/calistar/` and are **not** relicensed under this guide's CC BY 4.0 license. The PNG above is only a white-background raster rendering for this PDF. Calistar is an independent, open-source tool; use its models and worksheet under the project's own license and attribution terms.

Commercial alternative credit. The separately licensed, paid Califlower Calibration Tool Mk2 by Adam Meadows / Vector 3D is the official for-profit product commonly associated with this style of dimensional-and-skew calibration. It is not the source of the Calistar diagram above, and no Califlower files or instructions are reproduced in this guide.

The Kickstarter/Autodesk FDM assessment protocol is a useful tiered reference: it asks for separate X and Y measurements at multiple nominal sizes and explicitly compares the axis averages. Its published scoring conditions use a controlled PLA material; a PCTG run is still valuable as a machine-and-material characterization, but its score is not comparable to that PLA reference.

Benchy is an excellent diagnostic artifact, not a dimensional-compensation calculator. Its nominal `60 × 31 × 48 mm` envelope is a useful reference, but the hull/deck transition, wall sequencing, infill support, cooling, and line width make it the wrong sole basis for changing machine motion or a whole-model scale. Treat a localized hull line as a model/thermal diagnostic unless the same periodic or directional error appears on a separate artifact.

Classify the Error Model

Proportional material scale error

The error grows with feature size. Use per-filament shrinkage or scale compensation.

Nearly constant external offset

Different-sized external features are all too large or small by a similar absolute amount. Use contour compensation after confirming flow and PA.

Internal holes differ from external contours

Use hole compensation, precise-wall controls, polyholes, or a design tolerance after testing multiple hole sizes and orientations.

Bottom layers alone are enlarged

Use elephant-foot compensation, first-layer/Z correction, or bed-temperature changes. Do not scale the entire model.

X and Y disagree

Investigate belts, squareness, mechanics, axis-specific motion, cooling direction, and measurement method before applying one shared material value.

Opposite-signed X/Y error is a stop condition

If one axis is long while the other is short, do **not** apply a shared XY shrinkage value, global model scale, or a flow change. A common proportional correction cannot make `X = nominal + error` and `Y = nominal - error` both better. First run a multi-size or multi-point XY artifact, keep the material profile unchanged, and inspect belt path/tension, gantry squareness, motion repeatability, cooling direction, and the measurement method. Only then decide whether the result is a motion issue, skew, a persistent contour offset, or material behavior.

For example, a Benchy whose length is `+0.50 mm` from its nominal reference while width is `-0.50 mm` is evidence for an axis-specific follow-up test, not evidence for a `+/- 0.5 mm` slicer-scale edit. Record the signed readings; averaging their magnitudes would hide the diagnostic fact that the signs disagree.

Proportional Compensation Formula

For slicers that request the measured cooled percentage, including OrcaSlicer's documented shrinkage field:

```
measured percentage = measured dimension / nominal dimension * 100
effective correction scale = 100 / measured percentage
```

If a nominal `100.00 mm` feature measures `99.40 mm`:

```
measured percentage = 99.40%
correction scale = 100 / 99.40 = 1.006036
```

The slicer expands the model by about `0.6036%`.

Worked Field Example

A controlled `20.00 mm` cube measured:

```
X = 20.13 mm
Y = 20.13 mm
Z = 20.00 mm
```

The measured XY percentage was:

```
20.13 / 20.00 * 100 = 100.65%
```

The corresponding effective scale was:

```
100 / 100.65 = 0.993542
```

After entering `100.65%` as the XY measured percentage and retaining `100%` in Z, the confirmation cube measured exactly `20.00 x 20.00 x 20.00 mm`.

This result was produced in a controlled September 2026 experiment by William Tinney with analysis by Codex. It demonstrates the convention, not a universal material value. A small cube is a fast test; confirm the result on a larger artifact and a functional tolerance model.

Why a Value Above 100% Can Be Correct

The field name may say "shrinkage," but OrcaSlicer documents it as the percentage actually measured after cooling. If a printed feature is larger than nominal, the measured percentage is above `100%`, and the slicer scales it down.

Always verify the convention used by the exact slicer and version. Other tools may request a correction percentage rather than a measured percentage.

Never Fix Material Scale with Axis Steps

Axis steps or rotation distance describe the machine's motion. Changing them to fix one filament makes every other material and machine coordinate wrong. Use axis calibration only for a demonstrated motion error, and use filament shrinkage for material-specific proportional error.

Holes and Fits

Printed holes are influenced by polygon approximation, extrusion width, pressure, cooling, seam placement, bridge behavior, and the measurement axis. Test several diameters and orientations.

For mating parts, tune to the required clearance rather than an abstract perfect number. Keep separate acceptance classes for sliding, locating, press, and threaded fits.

Avoid Double Compensation

Do not manually scale a model when the filament profile already applies shrinkage unless the model needs an additional intentional change. Likewise, do not bake a calibration transform into a project and apply the same profile correction again.

With geometry credible, refine Seams and Surfaces.

Seams and Surface Quality

A seam is the repeated start and stop of a perimeter. It concentrates pressure, ooze, travel, retraction, wipe, and placement effects into one visible region. Treat it as a local transition problem, not automatic evidence that global flow is wrong.

Read the Seam in Two Parts

End of the previous perimeter

- Bulge: pressure may remain too high; PA may be low.
- Gap or thin finish: PA may be high, the seam gap may be large, or wipe may be excessive.

Start of the next perimeter

- Blob: ooze, over-prime, short travel, temperature, or inconsistent restart.
- Gap: excess retraction, delayed pressure recovery, large seam gap, or insufficient restart.

Long and short travels can produce different starts. Compare both before tuning one global restart value.

Tuning Order

1. Dry the filament and settle temperature.
2. Tune MVS, PA, and global flow.
3. Choose a repeatable seam location for diagnosis.
4. Tune retraction and wipe if travel artifacts remain.
5. Adjust seam gap or scarf behavior in small increments.
6. Validate around curves and sharp corners.

Aligned seams make diagnosis easier. Random seams can make cosmetic texture less concentrated but hide a systematic error.

Conventional Seam Controls

- seam position or painting;
- seam gap;
- wipe before or while retracting;
- retraction distance and speed;
- travel path and wall-crossing avoidance;
- perimeter order;
- start-point placement and corner preference.

Change one control at a time because several of these alter the same material transition.

Know When to Reset the Baseline

If several seam controls only move, lengthen, or multiply the disturbed region, stop changing seam values. Reconfirm the physical nozzle, selected tool, resolved line widths, layer height, PA, and final G-code. A false nozzle or tool baseline can make scarf and stagger experiments look active without making any of their numerical results valid.

Use aligned seams during diagnosis. Staggering can distribute one transition error into several wall regions and make cause and effect harder to read. A scarf can likewise spread an incorrect pressure or geometry transition instead of curing it.

The photographed FibreSeek seam reset shows this pattern: a broad disturbed region survived many cosmetic controls, then collapsed to one narrow seam when the actual `0.4 mm` nozzle and matching process geometry were restored.

Scarf Seams

A scarf seam spreads a layer transition over a distance instead of placing the entire start/stop event at one point. It can reduce a vertical ridge but adds variables such as scarf length, start height, conditional use, wall selection, and gap behavior.

Tune scarf settings only after ordinary pressure and flow are stable. A scarf can spread an incorrect transition without correcting its cause.

Surface Artifact Map

Surface evidence	First suspects
Repeating vertical line at one XY point	seam transition and placement
Bulge on every sharp corner	PA low, acceleration, wall order
Thin approach to corners	PA high or flow transition
Periodic vertical ripples after corners	ringing/input shaping/mechanics
Repeating fine vertical texture	VFA, motor/motion system, belt behavior
Random pits or bubbles	moisture, contamination, nozzle debris
Horizontal banding at regular Z spacing	Z motion, lead screw, temperature cycle
Gloss changes at feature transitions	speed/flow/temperature/cooling change
Rough top only	top flow, support below top, top speed/width/pattern
Wall roughness only at high speed	MVS, temperature, extrusion-rate transition

Top and Bottom Surfaces

Surface quality depends on more than flow:

- shell thickness;
- pattern and direction;
- top/bottom speed and acceleration;
- line width;
- overlap with perimeters;
- infill support;
- monotonic ordering;
- ironing, if used;
- lower-layer flatness.

Tune ironing last. It can hide top texture but introduces its own flow, spacing, speed, heat, and edge behavior.

Read repeated line-end defects as path evidence

When one edge of a top surface has a bead at nearly every line endpoint while the center has adequate coverage, do not immediately lower global flow. Inspect the sliced toolpath first:

1. Count the separate surface strokes and the non-extruding moves between them.
2. Compare those travel distances with the minimum retraction distance.
3. Check whether retraction and wipe are actually emitted in the G-code.
4. Compare line starts with line ends; a fat end and lean start indicates a pressure transition rather than too much material across the whole surface.

Short hops between adjacent lines often fall below the retraction threshold. The nozzle then stops, moves, and restarts pressure repeatedly along the same boundary. The resulting endpoint beads, hairs, or visible seam can survive an otherwise correct flow ratio and pressure-advance value.

Change the path topology before disturbing a validated material profile. A one-variable comparison between Monotonic Line and Monotonic is a useful test: the former avoids perimeter overlap but can leave visible seams, while the latter may trade those seams for more material interaction at the perimeter. Judge both the boundary and the center. If pattern choice does not solve the artifact, test retraction and wipe behavior separately. Do not combine a pattern, flow, line-width, and pressure-advance change in one trial.

Judge in Appropriate Light

Glossy and dark polymers exaggerate small ridges under flash. Use diffuse and glancing light, touch the surface, and compare a control. Decide whether the surface fails the intended use, not whether one photograph can reveal a line.

If the evidence changes with direction or steady wall speed, continue with Motion Diagnostics. Otherwise finish with Validation and Profile Release.

Validation and Profile Release

A calibration value is not finished when its test looks good. A profile is finished when the complete settings work together on representative parts and the evidence can be reproduced.

Minimum Validation Set

1. Broad surface coupon

Checks global and top-surface flow, line continuity, wall contact, and visible surface defects.

2. Dimensional artifact

Checks X, Y, and Z scale, external contours, several hole sizes, elephant foot, and at least one functional clearance.

Use more than one dimensional artifact when the symptom crosses categories. A Benchy is a strong geometry-and-surface diagnostic but is not a substitute for a multi-size dimensional stack, a multi-point XY/skew grid, or raw dial-gauge motion verification. Preserve signed X and Y readings rather than reporting only absolute error; opposite signs require an axis-specific investigation before any global compensation is accepted.

3. Geometry artifact

Checks bridges, overhangs, small towers, text, curves, corners, and travel stringing.

4. Representative production part

Uses the actual layer height, wall strategy, infill, support, speed, chamber, and build plate expected in service.

5. Mechanical specimen when needed

Validates layer adhesion or functional strength at the production temperature, flow rate, cooling, orientation, and environmental conditioning.

Acceptance Criteria

Define limits before printing. Examples:

No missing or discontinuous perimeter lines.
No open top-surface channels visible under diffuse light.
External dimensions within +/- 0.10 mm over 50 mm.
Specified pin passes the sliding-fit hole without force.
No corner lift greater than 0.20 mm.
No layer separation in the defined bend comparison.

Avoid criteria such as "looks good" without a reference or failure threshold.

Confirm the Operating Envelope

Validate the profile at the edges that production will use:

- lowest and highest normal layer heights;
- slow external walls and fastest infill;
- short and long layers;
- small and large bed footprints;
- longest expected print duration;
- enclosure and ambient extremes;

- tool changes or material changes when applicable.

A single cube cannot validate all of these.

Review the G-Code

Before a final release, inspect the sliced result for:

- selected physical tool and nozzle, not only the preset label;
- bed mesh and offset behavior;
- temperature transitions;
- fan commands and chamber behavior;
- PA or Linear Advance commands;
- flow and feature-specific overrides;
- actual volumetric-flow range;
- layer heights and line widths consistent with the installed nozzle;
- model scaling and compensation;
- unexpected auxiliary, fiber, or tool-change commands;
- unintended calibration-mode commands.

Where the machine exposes live state, compare the important emitted commands with the effective firmware values after printing begins. The last writer wins; an earlier start-G-code command can be replaced by a later slicer command.

For a networked multi-tool printer, extend this review to the complete job lifecycle. Confirm the uploaded file path, perform a real home before launch, and audit not only the start tool but every load, purge, cutter, tool-change, and unload command through the end block. Then verify the remote printer's active filename, selected tool, temperatures, PA, and job progress after it starts. A control request timing out does not establish its outcome; live state is the authoritative record.

Version the Profile

Record:

- profile name and semantic or dated revision;
- parent printer and process profiles;
- slicer and firmware versions;
- accepted values and their experiment records;
- known limits and provisional values;
- date, author, and reason for change.

Never overwrite the only known-good profile during experimentation. Duplicate it, mark the new copy `candidate`, and promote it after validation.

Keep tuned filament profiles machine-specific. Include the machine, tool/nozzle, purpose, and revision in the name or metadata, and restrict compatibility to the matching printer definition. Preserve the generic or supplier profile as an untouched reference.

A visually improved coupon is still a candidate. Promote it only after a clean project reproduces the slice and a representative functional part passes its dimensional, fit, surface, and strength requirements.

Regression Triggers

Run at least a short confirmation after:

- changing nozzle, hotend, extruder, or cooling duct;
- updating firmware or slicer behavior;
- switching material color, formulation, or lot;
- changing temperature, MVS, or PA materially;
- changing enclosure or chamber operation;
- observing a sudden quality change from a previously stable profile.

Release Decision

Use the profile release checklist. A released profile should state what it is optimized for: visual quality, balanced use, speed, strength, dimensional work, or a specific process. One profile does not need to win every tradeoff.

Material-Family Priorities

These notes describe common tendencies, not fixed settings. Formulations vary widely. Start with the manufacturer's current limits and safety information, then use the same calibration workflow.

PLA and PLA Blends

Priorities:

- avoid excessive chamber heat and heat creep;
- provide enough cooling for overhangs and small features;
- verify high-flow strength rather than judging appearance alone;
- retest filled, silk, matte, foaming, and impact-modified products separately.

Silk and filled PLA can require different flow, temperature, PA, and MVS from a plain PLA of the same brand.

PETG and PCTG

Priorities:

- dry thoroughly before blaming retraction for strings;
- avoid excessive nozzle squish on the first layer;
- balance cooling against layer bonding;
- bracket temperature high enough to find the true optimum at production flow;
- expect sticky buildup on the nozzle to create blobs and scars.

If a temperature tower keeps improving at the hottest section, the optimum has not been bracketed. Extend the range within manufacturer limits and include a strength comparison.

ABS and ASA

Priorities:

- stable enclosure and chamber conditions;
- controlled drafts and conservative cooling;
- bed adhesion and heat soak;
- dimensional shrinkage after extrusion is tuned;
- ventilation appropriate for the material.

Warping, corner lift, and layer splitting are usually thermal-management problems before they are flow problems.

Polycarbonate and PC Blends

Priorities:

- very dry filament;
- suitable high-temperature hardware and build surface;
- stable warm chamber when recommended;
- low uncontrolled cooling;
- strength validation at the intended flow rate.

Check the entire filament path and spool temperature rating before drying or printing at high temperature.

Nylon and Polyamide Blends

Priorities:

- strict moisture control and printing from a dry enclosure;
- nozzle wear for filled grades;
- dimensional conditioning after printing;
- creep, flexibility, and functional-fit validation;
- conservative MVS for highly filled formulations.

Nylon can change dimensions as it absorbs moisture. Define when dimensions are measured and the expected service humidity.

TPU and Other Flexible Materials

Priorities:

- low-resistance filament path and controlled extruder tension;
- conservative speed and volumetric flow;
- minimal necessary retraction;
- PA tuned specifically for the material and path;
- dimensional tests under relaxed, uncompressed conditions.

Hardness alone does not define print behavior. Different TPU chemistries with the same Shore rating can need different profiles.

Fiber-, Metal-, Wood-, and Mineral-Filled Materials

Priorities:

- manufacturer-required nozzle hardness and minimum diameter;
- nozzle wear tracking;
- reduced clog risk and suitable retraction;
- MVS testing rather than relying on the unfilled base polymer;
- ventilation and handling appropriate to the filler;
- dimensional validation because fillers alter shrinkage and anisotropy.

The base-polymer profile is only a starting point.

Support and Soluble Materials

Priorities:

- moisture control;
- reliable tool-change purge and priming;
- interface temperature compatibility;
- idle-temperature and ooze management;
- storage and post-processing safety.

Validate the complete material pair, not each filament in isolation.

OrcaSlicer and TinmanX1 Settings Reference

This part of the guide translates the process-profile screens into practical decisions. It is not a replacement for calibration and it is not a rewritten copy of the OrcaSlicer Wiki. Each setting group answers three questions:

1. **What does this group control?**
2. **When should you use it?**
3. **What result should you be trying to produce?**

The control maps name the settings found in the interface, but the advice is organized around print evidence. If a part does not show the symptom described for a group, leave that group alone.

Version Basis

This edition was checked against the following fixed references on 2026-09-21:

Reference	Version or revision	What it establishes
OrcaSlicer application source	`v2.4.2`, commit `8500fcdccaa10b5099ac20d252af3a7c560046f1`	The upstream 2.4.2 process tabs and controls
OrcaSlicer Wiki	commit `f132cbf1c1e48573d229d1783ee471e61e50aaf3`, dated 2026-09-17	Current upstream terminology and supporting context
TinmanX1 source	upstream-compatible version `2.4.2`, commit `cf0cabb04146c4ff9f21b963517717acfec66055`	The TinmanX1 UI, fiber planner, Strength Lens, and Wave Overhang extensions
Installed TinmanX1 application	`CFBundleShortVersionString 2.4.2`	The application version visible to the user

TinmanX1 keeps the OrcaSlicer `2.4.2` version number because that is its compatibility base. The TinmanX1 commit is therefore the more precise identity for custom features. A later TinmanX1 build may show the same application version while containing newer custom work.

Official download pages:

- OrcaSlicer releases
- TinmanX1 releases

The links intentionally point to each project's latest public release. Compare the downloaded version with the fixed documentation baseline above before assuming every screen and option will be identical.

Difference Labels

- **Shared 2.4.2:** the group and surveyed controls are present in both OrcaSlicer 2.4.2 and the TinmanX1 baseline.
- **TinmanX1 addition:** the group or control is part of TinmanX1 and is not in upstream OrcaSlicer 2.4.2.
- **Profile contract:** the field describes hardware or saved planner data. It normally belongs to a validated printer, filament, or process preset rather than routine user tuning.
- **Experimental:** preview the sliced paths and validate on a disposable test before using the feature on a valuable print.

These labels describe the stated source revisions, not every older or future release. A setting added to OrcaSlicer after 2.4.2 may later become shared.

Scope

The following chapters cover every process-profile tab and group in the reference builds:

- Quality
- Strength
- Speed
- Support
- Multimaterial
- Others
- TinmanX1 fiber and strength tools

The last chapter also explains the continuous-fiber fields in the TinmanX1 filament and FibreSeek printer profiles. Those fields are outside the ordinary process tabs, but they are included because a complete fiber setup depends on all three profile layers.

How to Use the Reference

Start with the symptom, not the menu. Read the group-level **What**, **Use it when**, and **Desired result** statements. Change the smallest relevant control, slice again, and inspect the preview before printing. Preserve a control sample and avoid changing material flow, geometry compensation, and path generation in the same experiment.

Expert mode exposes more controls but does not make them equally useful. Zero often means "inherit," "automatic," or "disabled," depending on the field. Confirm the field's units and preview the effect instead of assuming that zero has one universal meaning.

Quality Tab

Version note: every group below is **Shared 2.4.2**. Source comparison found no TinmanX1-only Quality controls in the stated baseline. TinmanX1 adds related tools elsewhere, especially Strength Lens, fiber planning, and Wave Overhangs.

The Quality tab controls how the slicer turns model geometry into printable roads. It can improve a surface only when the defect matches the kind of path this tab changes. Do not use it to disguise wet filament, a loose machine, a clog, or a melt-flow limit.

Layer Height

What: sets the vertical thickness of normal and first layers. It changes detail, staircase visibility, print time, and the amount of plastic demanded by each road.

Use it when: curves look stepped, tiny Z details disappear, the first layer needs a different thickness, or the chosen height is poorly matched to the nozzle.

Desired result: the coarsest layer height that preserves the required geometry, with a first layer that adheres without excessive squish.

- `Layer height` controls ordinary layers.
- `Initial layer height` trades first-layer detail for tolerance of small bed and Z errors. A thicker value is not permission to ignore a bad mesh or incorrect Z offset.

Line Width

What: sets the planned width of roads for each feature. Width changes the number of paths, contact between adjacent roads, detail resolution, flow demand, and wall thickness.

Use it when: a particular feature is weak, poorly closed, too coarse, or visibly different from the rest of the print after global flow is calibrated.

Desired result: widths that the nozzle can deposit consistently, with joined roads and no unnecessary loss of detail.

- `Default line width` is the fallback.
- `Initial layer line width` changes bed contact.
- `Outer wall line width` trades fine geometry against a broader, more forgiving shell road.
- `Inner wall`, `sparse infill`, and `internal solid infill line width` govern structural roads.
- `Top surface line width` directly affects top closure and visible line scale.
- `Support line width` and `bridge line width` affect unsupported structures.

Change width before feature-specific flow when the problem is road geometry. Change flow when road geometry is sensible but deposited volume is wrong.

Seam

What: controls where perimeter loops begin and end and how pressure is handed across that transition.

Use it when: one vertical scar, blobs, pits, hairs, or start/stop defects repeat at the seam while the rest of the wall is sound.

Desired result: a predictable transition that is either hidden in a chosen location or distributed without creating widespread marks.

- `Seam position` chooses the placement strategy: nearest, aligned, rear, or random. Aligned is best for diagnosis.
- `Staggered inner seams` separates inner-wall starts from the visible outer seam.

- `Seam gap` deliberately leaves a small end gap; too much creates a notch and too little can leave a ridge.
- `Wipe speed`, `wipe on loops`, and `wipe before external loop` control motion around the stop/start event.
- Scarf controls spread the transition over distance: `scarf joint type`, conditional use, threshold angles, speed, start height, full-loop use, minimum length, steps, inner-wall use, and flow ratio.

Tune pressure advance and global flow first. A scarf can distribute a bad transition, but it cannot make an incorrect extrusion system correct.

Precision

What: changes how closely sliced paths follow geometry and applies targeted dimensional corrections.

Use it when: measured errors have a known model: constant contour offset, hole-only error, bottom-layer flare, curve faceting, or lost narrow geometry.

Desired result: cooled parts meet the intended dimensions without applying one correction to unrelated errors.

- `Slice gap closing radius` closes very small gaps in sliced regions. Excessive values can merge features that should remain separate.
- `Resolution` controls curve simplification; smaller values keep more points but produce larger, busier paths.
- `Arc fitting` replaces suitable short segments with arcs when the output and firmware support them.
- `X-Y contour compensation` offsets outside contours. It is for a nearly constant external error, not proportional material shrinkage.
- `X-Y hole compensation` offsets internal holes separately.
- `Elephant foot compensation`, layer count, and density target only the enlarged bottom region.
- `Precise wall` improves wall placement where line-center approximations would bias thickness.
- `Precise Z height` adjusts layer planning to finish closer to the requested height.
- Polyhole controls convert eligible round holes to printer-friendly polygons and define the size threshold and orientation.

See Dimensional Accuracy and Tolerances before using compensation.

Ironing

What: makes an extra low-flow pass over selected solid top surfaces to press and redistribute plastic.

Use it when: an already closed, supported top needs a smoother cosmetic finish. Do not use it to repair open channels, pillowing, or sparse support below the roof.

Desired result: a more uniform sheen and reduced ridge height without edge bulges, drag marks, or heat damage.

- `Ironing type` selects which surfaces receive the pass.
- `Pattern` chooses rectilinear or concentric motion.
- `Flow` supplies the small amount of material used to fill valleys.
- `Spacing` sets pass density; tighter spacing increases smoothing and heat.
- `Inset` keeps the tool away from perimeter edges.
- `Angle` and `fixed angle` control direction.
- `Ironing speed` is on the Speed tab and must be judged with flow and spacing.

Tune ironing last.

Z Contouring

What: varies path height within a layer to represent sloped top geometry with less visible stair stepping.

Use it when: a shallow, upward-facing slope is visibly terraced and ordinary layer-height reduction is too slow or still leaves objectionable steps.

Desired result: a smoother slope with stable extrusion and no collisions or thin unsupported ridges.

- `Z contouring` enables the strategy.
- `Minimize wall height angle` limits where variable-height walls are attempted.
- `Minimum Z height` protects against impractically thin vertical changes.
- `Do not alternate fill direction` keeps the fill direction consistent when alternating it would harm the contoured surface.

This changes three-dimensional path geometry. Inspect the preview carefully and validate clearance before production use.

Wall Generator

What: chooses how the slicer fits wall roads into changing thickness. Classic uses mostly fixed widths; Arachne varies width to fill narrow or tapered areas.

Use it when: thin features vanish, wall thickness changes create gaps, or variable-width walls themselves create an uneven visible finish.

Desired result: complete walls with stable widths, preserved detail, and no needless width oscillation.

- `Wall generator` chooses Classic or Arachne.
- `Wall transition angle`, `filter deviation`, and `transition length` decide how readily Arachne changes wall count and width.
- `Wall distribution count` controls how many neighboring walls share a width transition.
- `Minimum wall width` and `first-layer minimum wall width` reject roads too narrow to print reliably.
- `Minimum feature size` and `minimum wall length` suppress tiny fragments.
- `Maximum wall resolution` and `maximum wall deviation` simplify highly detailed wall paths while limiting geometric error.

For surface finish, start with the generator choice and preview the outer wall. Classic can look more uniform on simple boxes; Arachne often fills variable geometry better. Neither corrects a wet filament or unstable extrusion system.

Walls and Surfaces

What: controls wall order, flow by feature, travel around walls, and special handling of small or top regions.

Use it when: walls are dimensionally sound but show order-related marks, small top areas behave differently from broad ones, or one feature needs a small flow correction after global calibration.

Desired result: a clean outer shell, closed surfaces, minimal travel scars, and feature-specific flow that does not disturb the validated global profile.

- `Wall printing order` controls whether outer or inner walls are printed first. Outer-first can protect dimensions; inner-first often gives the visible wall more support. Inner-outer-inner is a deliberate compromise.
- `Print infill first` changes what supports the wall and can make infill marks more visible through a thin shell.
- `Wall loop direction` can move seam and overhang behavior relative to machine cooling and geometry.

- `Print flow ratio`, top, bottom, first-layer, wall, overhang, sparse-infill, solid-infill, gap-fill, support, and support-interface flow ratios are targeted multipliers. Enable the additional ratios only when evidence identifies a feature-specific error.
- `Only one wall` options reduce walls on the first layer or top surfaces where extra walls would crowd a narrow region.
- `Minimum top-surface width` decides what is treated as a visible top.
- `Avoid crossing walls` and `maximum detour` trade travel distance for fewer scars across visible shells.
- `Small-area flow compensation` and its model reduce overfill caused by very short extrusion segments. Treat the model as an advanced calibration curve, not a cosmetic randomizer.

This group is often relevant to surface finish, but change path order or the specific feature before disturbing global flow.

Bridging

What: controls roads laid across air or sparse internal support.

Use it when: unsupported spans sag, separate, curl, or become too dense while ordinary walls print correctly.

Desired result: straight, attached bridge lines with acceptable sag and a usable surface above them.

- External and internal `bridge flow` change material per bridge road.
- External and internal `bridge density` change spacing between roads.
- `Thick bridges` selects the older thick-road behavior when needed.
- `Extra bridge layer` adds a reinforcing layer above chosen bridges.
- `Filter small internal bridges` decides whether short internal spans receive bridge treatment.
- `Counterbore-hole bridging` chooses how flat-bottom holes are closed.

Bridge speed is on the Speed tab and cooling is in the filament profile. Tune the three together with separate tests.

Overhangs

What: changes wall generation for partially unsupported edges.

Use it when: steep walls curl, droop, lose their edge, or need geometry adaptation before ordinary support is added.

Desired result: attached overhang roads and a stable edge without needless geometry distortion.

- `Detect overhang wall` allows feature-specific speed and flow behavior.
- `Make overhang printable` modifies difficult geometry; angle and hole-size limits determine where that modification is allowed.
- `Extra perimeters on overhangs` creates more local support.
- `Reverse on even`, `reverse internal only`, and the reversal threshold change direction between layers to balance one-sided deposition effects.

Overhang speed belongs to the Speed tab, while fan and layer-time controls belong to the filament profile. TinmanX1's experimental Wave Overhang system is documented separately because it generates a different supportless path family.

Strength Tab

Most of this tab is **Shared 2.4.2**. TinmanX1 adds the **Strength Lens** group at the top; OrcaSlicer 2.4.2 begins with Walls. Strength settings affect the amount and arrangement of material, but settings alone do not prove strength. Validate critical parts with representative loads and failure tests.

Strength Lens

Difference: TinmanX1 addition. OrcaSlicer 2.4.2 does not have this pre-slice orientation aid.

What: colors the prepared model according to a simplified directional strength model before slicing.

Use it when: deciding how to orient a part relative to an expected load, especially when layer separation or continuous-fiber direction matters.

Desired result: an orientation that keeps the important load in stronger printed directions and moves weak layer-normal loading away from critical features.

- `Strength Lens` turns the view on.
- `Material model` selects the assumption: filament preset, isotropic, ordinary anisotropic FDM, or continuous fiber.
- `Load axis` chooses Auto, X, Y, or Z for the direction being evaluated.

The lens is a comparative visualization, not finite-element analysis or a strength certification. See TinmanX1 fiber and strength tools for the model limits.

Walls

What: controls the number and continuity of perimeter shells.

Use it when: a part needs better bending, impact, sealing, screw retention, or exterior robustness. For many real parts, an extra wall is more useful than the same plastic added as sparse infill.

Desired result: a continuous shell thick enough for the load without crowding small features or adding unnecessary mass.

- `Wall loops` is the primary shell-thickness control.
- `Alternate extra wall` adds reinforcement on alternating layers, improving shell-to-infill connection with less material than another wall everywhere.
- `Detect thin walls` preserves narrow model features Classic walls might omit; with Arachne, inspect the preview because variable-width walls already solve many thin-feature cases.

Top and Bottom Shells

What: controls the solid roofs and floors that close the part and transfer load across sparse infill.

Use it when: top surfaces pillow or remain open, bottom skins are fragile, or a face needs greater puncture and bending resistance.

Desired result: fully closed skins with enough total thickness to bridge the infill and survive the intended load.

- `Top shell layers` and `bottom shell layers` set counts.
- `Top shell thickness` and `bottom shell thickness` enforce physical thickness even when layer height changes. The slicer uses whichever requirement demands more layers.
- `Top surface density` and `bottom surface density` control how fully the visible surface is filled.
- `Top surface pattern` and `bottom surface pattern` change path direction, continuity, and appearance. Monotonic families improve visual ordering; concentric follows outlines; decorative patterns trade speed and uniformity.
- `Top/bottom infill-wall overlap` joins solid skins to the shell. Too little leaves a gap; too much produces a raised perimeter band.

Do not tune a rough top with shell count alone. First determine whether the problem is inadequate support, wrong top flow, path endpoints, or heat.

Infill

What: creates the internal load path and support structure between shells.

Use it when: the part needs a different stiffness-to-weight tradeoff, the roof needs better support, or the load has a known direction.

Desired result: enough connected internal structure to support skins and carry the expected load without wasting material or creating excessive nozzle crossings.

- `Sparse infill density` controls how much of the interior is occupied.
- `Sparse infill pattern` selects the load path. Rectilinear families are fast; gyroid and cubic families distribute load in more directions; lightning is mainly roof support, not general strength.
- `Infill direction` and the sparse rotation template orient or sequence layers.
- `Symmetric infill Y axis` mirrors orientation where part symmetry matters.
- `Infill-wall overlap` joins sparse infill to shells.
- `Infill anchor` and `maximum anchor length` control how far infill is tied into perimeters.
- `Internal solid infill pattern`, direction, and rotation template control buried solid regions.
- `Gap fill target` and `filter tiny gap fill` decide where narrow leftover spaces receive short roads. Excessive gap fill can create heat and blobs.
- `Extra solid infill` adds selected reinforcement through modifiers or painted regions.
- `Fill multiline` and the pattern-specific controls tune specialized patterns: optimized gyroid, locked-zag skin/skeleton density and widths, cross-hatch shift, lateral-lattice angles, lateral-honeycomb angle, and lightning branch angles.

Pattern-specific controls do nothing unless their matching pattern is active. Change them only after the broad pattern and density are justified by the load.

Advanced Strength

What: changes bridge orientation, combines infill layers, and adds solid material where shell continuity would otherwise be weak.

Use it when: preview inspection reveals a directional bridge problem, tiny sparse regions, inefficient infill on every layer, or thin sloped shells that are not receiving enough support.

Desired result: continuous load paths and adequately supported shells with no unnecessary solid mass or overly tall infill roads.

- `Align infill direction to model` makes orientation follow model coordinates instead of the plate, useful when rotating several copies of a directional part.
- `Bridge angle`, `internal bridge angle`, and `relative bridge angle` override automatic span direction. Use only after previewing the anchors.
- `Minimum sparse infill area` converts tiny sparse islands to solid fill.
- `Infill combination` prints sparse infill less often using taller roads; `maximum layer height` caps those roads. This saves time but can reduce local support and exceed flow limits.
- `Detect narrow internal solid infill` handles thin buried solids that a normal pattern cannot fill cleanly.
- `Ensure vertical shell thickness` adds material below sloped surfaces so the requested shell remains physically thick rather than merely having a fixed number of layers.

Speed Tab

All groups in this chapter are **Shared 2.4.2**. A speed target is a request, not a guarantee: acceleration, path length, cooling, and maximum volumetric flow may prevent the machine from reaching it.

First Layer Speed

What: controls extrusion and travel while adhesion is being established.

Use it when: the first layer drags, curls, fails to join, or is disturbed by fast travel even though Z offset and plate preparation are correct.

Desired result: continuous, well-attached roads with clean direction changes and no nozzle pickup.

- `Initial layer speed` covers first-layer walls and ordinary extrusion.
- `Initial layer infill speed` sets broad first-layer fill independently.
- `Initial layer travel speed` controls non-extruding moves near fresh roads.
- `Number of slow layers` ramps toward normal speed instead of changing abruptly after layer one.

Other Layers Speed

What: assigns speed targets by feature role.

Use it when: one role has a different finish, dimensional, cooling, or flow requirement from the rest of the part.

Desired result: visible and critical features receive conservative motion, while hidden, well-supported features use available throughput safely.

- `Outer wall speed` strongly influences visible finish and dimensional consistency.
- `Inner wall speed` can be faster, provided it remains within melt and motion limits and does not disturb the outer wall.
- `Small perimeter speed` and its size threshold slow short loops that cannot settle thermally or dynamically.
- `Sparse infill speed` is usually throughput-oriented.
- `Internal solid infill speed` must still provide a stable base for top skins.
- `Top surface speed` trades time for visible consistency and closure.
- `Gap infill speed` protects short, pressure-sensitive roads.
- `Ironing speed` works with ironing flow and spacing.
- `Support speed` and `support interface speed` separate disposable structure from the layer touching the model.

If a defect begins only above a certain speed, confirm maximum volumetric flow before lowering every feature speed.

Overhang Speed

What: slows partially unsupported walls by degree and assigns external and internal bridge speeds.

Use it when: ordinary walls are clean but overhang edges curl or bridges sag.

Desired result: each new road remains attached to the previous layer and has time to cool before the next pass.

- `Slow down for overhangs` enables percentage-based overhang bands.
- The four overhang speeds cover increasingly unsupported portions of a road; zero means use the normal wall behavior for that band.

- `Slow down for curled perimeters` reduces speed where the slicer predicts a collision-prone curled edge.
- External and internal `bridge speed` can differ because visible spans and buried spans have different priorities.

Cooling and nozzle temperature can dominate this test. Keep them fixed while comparing speeds.

Travel Speed

What: sets non-extruding movement speed.

Use it when: travel time contributes to ooze, stringing, or heat marks, or the machine becomes noisy and inaccurate during rapid repositioning.

Desired result: prompt movement between features without skipped motion, frame excitation, or disturbance of fresh plastic.

High travel speed can reduce ooze time but only if acceleration and mechanics can support it. A short move may never reach the requested value.

Acceleration

What: controls how quickly speed changes for each feature role.

Use it when: ringing, rounded corners, weak start/stop behavior, or excessive print time comes from transitions rather than steady speed.

Desired result: the highest acceleration that preserves placement, corner quality, extrusion response, and mechanical reliability.

- `Normal printing acceleration` is the fallback.
- Outer wall, inner wall, bridge, sparse infill, internal solid infill, top surface, first layer, first-layer travel, and travel accelerations provide role-specific limits.
- `Acceleration to deceleration` and its factor limit abrupt transition energy where supported by the selected firmware and G-code flavor.

Pressure advance must be calibrated for the actual acceleration regime. Do not use low acceleration to conceal badly tuned extrusion pressure.

Junction Deviation

What: requests a cornering model based on allowed path deviation rather than a direct instantaneous speed-change number.

Use it when: the firmware and G-code flavor support junction deviation and corner behavior needs a profile-level limit.

Desired result: controlled corner speed without visible ringing or excessive rounding.

Do not tune junction deviation and classic jerk simultaneously. Confirm which model the firmware actually consumes.

Jerk (XY)

What: supplies feature-specific corner-transition limits for firmware that uses classic jerk-style controls.

Use it when: supported firmware shows ringing, harsh motion, or rounded corners that depend on feature role.

Desired result: crisp but mechanically calm direction changes.

- Default, outer-wall, inner-wall, infill, top-surface, first-layer, first-layer-travel, and travel jerk targets are available.

Values and meanings are firmware-specific. A number copied from another motion system is not a calibration.

Advanced Speed

What: limits how quickly commanded extrusion rate may change along a path.

Use it when: a high-flow transition creates gloss bands, pressure shocks, or inconsistent width even though maximum volumetric flow and pressure advance are already calibrated.

Desired result: smoother extrusion-rate transitions without slowing the entire model more than necessary.

- `Maximum volumetric extrusion rate slope` limits rate-of-change.
- `Slope segment length` sets the distance over which the limiter evaluates the change.
- `External perimeter only` confines smoothing to visible walls when internal throughput need not be limited.

This is an advanced transition tool, not a replacement for the filament's maximum volumetric speed.

Use Motion Diagnostics when the question is whether texture comes from acceleration, input shaping, VFA, or hidden geometry rather than a single feature-speed setting.

Support Tab

Ordinary support, raft, interface, advanced, and tree groups are **Shared 2.4.2**. The Wave groups are **TinmanX1 additions** and **experimental**. Support tuning is a compromise among underside quality, removability, print time, material, and the risk of a failed unsupported feature.

Support

What: decides whether support is generated, where it is allowed, and which broad geometry style it uses.

Use it when: a feature cannot be bridged or overhung reliably in its current orientation.

Desired result: the least support that makes the model succeed, with stable columns and reachable removal paths.

- `Enable support` turns generation on; automatic and manual support types decide whether angle rules or painted/enforced regions drive placement.
- `Style` chooses normal grid/snug or tree-family behavior.
- `Threshold angle` and `threshold overlap` control how unsupported a region may be before support appears.
- `Initial layer density` and `initial layer expansion` stabilize the support's own footprint.
- `On build plate only` prevents support from starting on the model.
- `Critical regions only` limits support to regions judged most likely to fail.
- `Ignore small overhangs` suppresses tiny, often self-supporting islands.

Orient the part before increasing support. A better orientation can improve strength and remove more support than any density adjustment.

Raft

What: creates sacrificial layers under the entire supported footprint.

Use it when: the part or support needs a stable, planar base that a brim cannot provide, or the machine/material combination makes first-layer recovery more important than bottom finish.

Desired result: a secure print that separates from the raft without tearing the part or leaving a badly distorted bottom.

- `Raft layers` controls raft thickness and rigidity.
- `Raft contact distance` controls release versus underside support. Smaller is tighter and cleaner underneath but harder to remove.

Support Filament

What: assigns material slots to support base and interface roles.

Use it when: a multi-material system can use a soluble, breakaway, or non-bonding interface material.

Desired result: reliable tool changes and an interface that supports the model yet releases or dissolves as intended.

- `Support filament` selects the base material.
- `Support interface filament` selects the contact material.
- `Avoid interface filament for base` preserves expensive or slow interface material for contact layers only.

Validate material compatibility; some pairs bond too strongly, too weakly, or at incompatible temperatures.

Support Ironing

What: smooths the top of support before the model is printed on it.

Use it when: support-interface texture transfers to a broad underside and ordinary interface spacing cannot provide a consistent platform.

Desired result: a flatter support roof without fusing the interface to the model.

- Enable, pattern, flow, and spacing work like ordinary ironing but apply to support. Heat buildup and excessive flow can make removal worse.

Wave Overhangs

Difference: TinmanX1 addition, experimental. OrcaSlicer 2.4.2 does not have Wave Overhangs.

What: replaces selected unsupported regions with wave-shaped paths intended to grow outward from supported material.

Use it when: conventional supports are undesirable and the feature is a candidate for carefully previewed supportless construction.

Desired result: connected wave paths that remain anchored and leave only small residual regions for ordinary support.

- `Use wave overhangs` enables the generator.
- `Use wave overhangs instead of bridges` extends it to spans that would otherwise use straight bridge roads.
- `Support remaining areas` keeps ordinary support under regions the wave planner cannot cover.

Always inspect the preview. Leave ordinary support available until the wave coverage has been validated for the exact part and material.

Wave Detection

What: decides which overhang regions qualify and how aggressively the planner grows them.

Use it when: waves appear on easy geometry, fail to reach a useful region, or consume excessive computation and path complexity.

Desired result: only genuinely difficult, sufficiently large regions are selected and solved in a reasonable number of iterations.

- Minimum angle and length reject easy or tiny candidates.
- Maximum iterations caps planning work and outward growth.

Wave Pattern

What: shapes and spaces the generated wave roads and determines how they meet ordinary perimeters.

Use it when: previewed waves lack support, leave gaps, overfill edges, or start in poor locations.

Desired result: continuous, evenly supported roads with a controlled joint to the real model boundary.

- Pattern selects monotonic, zigzag, or smart path ordering.
- Seam mode chooses alternating, aligned, or deterministic random starts.
- Perimeter count retains ordinary walls at the edge.
- Line spacing and spacing mode control wave density and progression.
- Perimeter overlap controls attachment to normal walls.
- Minimum width and minimum new area suppress fragments.
- Flow per path length controls deposited wave volume directly.

Wave Fringe Reinforcement

What: limits small reinforcing wave regions near the boundary between real geometry and generated cover.

Use it when: fringes either fail to support an edge or spread excessive material beyond the useful contact area.

Desired result: enough local reinforcement to anchor the fringe, with no large artificial shelf.

The cover-to-real ratio, maximum cover area, and contact-over-cap compensation are interacting safety bounds. Change one at a time and inspect the path area, not just the numeric value.

Wave Corner Reinforcement

What: increases support near corners where uniform wave spacing can leave a weak or abrupt transition.

Use it when: preview or test prints show corner droop, separation, or a visible density step.

Desired result: a gradual, locally denser corner region without a blob.

- Enable turns on corner tapering.
- Corner spacing, taper distance, and angle threshold set its density, reach, and eligibility.

Wave Motion

What: assigns extrusion, perimeter, travel, and end-retraction behavior to wave paths.

Use it when: geometry is correct in preview but physical waves detach, vibrate, ooze, or mark their endpoints.

Desired result: calm deposition that keeps each wave attached while moving cleanly between regions.

- Wave print, perimeter, and travel speeds separate the motion roles.
- End retraction controls pressure at the finish of a wave route.

Wave Cooling

What: applies wave-specific fans, nozzle temperature, and minimum timing.

Use it when: waves remain soft and curl, or cool so aggressively that they do not bond.

Desired result: each path is firm enough to support the next while still bonding to its anchor.

- Part and auxiliary fan targets control airflow.
- Nozzle temperature controls bonding and sag.
- Minimum wave and layer times prevent a rapid return to soft material.

Wave Floor Layers

What: builds stabilizing layers under the final wave-supported surface.

Use it when: the top of the generated support needs a more uniform platform or gradual speed/cooling transition.

Desired result: a stable floor that supports the model without becoming a fused mass.

- Floor layer count sets depth.
- Hilbert use, layer count, and density change floor topology.
- Floor print/perimeter speed, speed ramp, and fan targets control the thermal transition into the model.

Wave Debug

What: emits diagnostic G-code information for development and path audit.

Use it when: reproducing or reporting a planner problem, not as routine print tuning.

Desired result: enough trace evidence to explain a generated path without changing production geometry unintentionally.

Advanced Support

What: controls the air gaps, patterns, interfaces, clearance, and layer planning that determine support quality and release.

Use it when: the broad support style is correct but the underside, removal, clearance, or support stability is not.

Desired result: a continuous interface close enough to support the model, far enough to release, and clear of visible sidewalls.

- Top and bottom Z distances control vertical separation.
- Support wall loops and base pattern/spacing control base stiffness.
- Pattern angle rotates the support structure.
- Top and bottom interface layers, pattern, and spacing control the contact roof and floor.
- Support expansion grows or shrinks normal support laterally.
- Object XY distance protects sidewalls; first-layer gap protects the model's first-layer region.
- `Do not support bridges` trusts bridge paths instead of placing support below.
- `Maximum bridge length` limits that trust.
- `Independent support layer height` lets support use its own vertical schedule.

Tune Z distance in layer-height-aware increments. A distance that is not representable by the layer plan may not behave as the decimal suggests.

Tree Supports

What: controls branch tips, spacing, density, diameter, angle, and roots for tree-style support.

Use it when: support must reach isolated overhangs while avoiding much of the model surface.

Desired result: branches that are stiff enough to survive motion, narrow enough to remove, and able to reach the target without collision.

- Tip diameter controls contact size.
- Branch distance and top rate control how densely tips merge.
- Branch diameter and diameter angle control trunk growth toward the bed.
- Branch angle and preferred slow angle control reach versus stability.
- Organic variants apply to the organic style.
- Auto brim and brim width stabilize the roots.

Multimaterial Tab

All groups in this chapter are **Shared 2.4.2**. TinmanX1's continuous-fiber system is not configured through these ordinary color/material-change controls; see TinmanX1 fiber and strength tools.

Prime Tower

What: creates a sacrificial structure used to restore pressure, purge color or material, and stabilize tool changes.

Use it when: multi-material transitions produce missing starts, contamination, unstable pressure, or unreliable nozzle readiness.

Desired result: clean, repeatable transitions with the smallest tower that remains mechanically stable.

- Enable controls tower creation.
- Skip points and interface features tune path continuity around changes.
- Interface cooldown controls tower cooling during transition work.
- Framework, width, prime volume, brim width, and infill gap control tower structure and purge capacity.
- Rotation changes footprint orientation.
- Bridging distance, purge-line spacing, extra flow, and maximum purge speed control how purge roads are deposited.
- Wall type, cone angle, rib length/width, and wall fillet tune stability.
- `No sparse layers` avoids a weak intermittent tower.
- Single-extruder multimaterial priming enables startup behavior for that hardware class.

Inspect tower paths and confirm bed clearance. A tower that is too small can fail late and endanger an otherwise good part.

Filament for Features

What: assigns filament slots to outer walls, inner walls, sparse infill, internal solids, top surfaces, bottom surfaces, and the tower.

Use it when: different materials or colors intentionally serve different visual or structural roles.

Desired result: each feature uses the intended compatible material, with tool changes and purge volume that are justified by the design.

Do not assume that a stronger filament automatically produces a stronger mixed part. The interface between materials may govern failure.

Ooze Prevention

What: manages idle-tool temperature and preheating before the next use.

Use it when: an inactive nozzle drools, contaminates the model, or returns too cold and under-extrudes.

Desired result: the idle tool stays cool enough to resist ooze and reheats early enough to extrude correctly at its next path.

- Enable turns the strategy on.
- Standby temperature delta lowers the idle target.
- Preheat time and steps shape the return to printing temperature.

Avoid temperature changes outside the material's safe processing range.

Flush Options

What: diverts otherwise wasted purge material into internal features or designated objects.

Use it when: color purity and material compatibility allow purge to become part of the print.

Desired result: reduced waste with no visible contamination, weak interface, or incorrect support behavior.

- Flush into infill hides compatible purge inside sparse structure.
- Flush into objects uses specifically designated purge objects.
- Flush into support uses disposable support, provided the material remains suitable for support and interface roles.

Advanced Multimaterial

What: creates interlocking geometry between material regions and controls segmented-interface behavior.

Use it when: adjacent materials need more mechanical keying than a flat boundary provides.

Desired result: a larger, mechanically interlocked interface that does not break through visible surfaces or overwhelm small geometry.

- Interlocking beam enables the strategy.
- Interface shells add boundary structure.
- Segmented-region width and depth set where segmentation applies.
- Beam width, orientation, layer count, depth, and boundary avoidance define the mechanical key.

Use test coupons. Chemical compatibility and temperature remain decisive even when geometry is interlocked.

Others Tab

The groups below are **Shared 2.4.2**. They collect adhesion aids, special slicing modes, texture, output controls, scripts, and notes. Several can change machine behavior far beyond appearance, so preview and validate them carefully.

Skirt

What: prints one or more lines around the part without attaching them to it.

Use it when: the nozzle needs priming, first-layer behavior needs a visual check, or a simple draft shield is useful.

Desired result: stable extrusion before the part starts, with no wasted height or collision risk.

- Loops and minimum extrusion length control priming quantity.
- Type chooses combined or per-object skirts.
- Distance and start angle place the path.
- Speed and height control deposition and vertical extent.
- Draft shield and single-loop-after-first-layer turn the skirt into a thermal barrier.

Brim

What: attaches sacrificial first-layer roads to the part or selected corners to increase bed contact.

Use it when: corners lift or a narrow footprint needs more holding force.

Desired result: the part stays flat and the brim removes without damaging the edge.

- Type chooses automatic, ears, painted, outer, inner, both, or none.
- Width controls adhesion area.
- Object gap trades holding force for removability.
- Flow ratio adjusts brim deposition independently.
- Elephant-foot outline use makes the brim follow the compensated first-layer contour.
- Combine brims joins nearby objects.
- Ear angle and detection length decide which corners receive ears.

A brim cannot correct oil, debris, a bad Z offset, or severe enclosure drafts.

Special Mode

What: changes how geometry is interpreted and how objects or layers are ordered.

Use it when: the model has unusual mesh topology, needs sequential printing, or is intentionally a single-wall spiral.

Desired result: a valid, collision-free toolpath whose special behavior is visible and understood in preview.

- `Slicing mode` chooses regular, even-odd, or close-holes interpretation.
- `Print sequence` chooses by-layer or by-object printing.
- `Print order` chooses the intra-layer object order.
- `Spiral vase` removes ordinary seams by continuously raising a single wall.
- Spiral smoothing, maximum XY smoothing, and start/finish flow ratios shape the transition into and out of the spiral.
- `Timelapse` changes park/motion behavior for image capture.
- `Wrapping detection` enables the supported wrap/clump detection workflow.

Sequential printing requires verified toolhead and gantry clearance. A safe bed layout is part of the setting, not an afterthought.

Fuzzy Skin

What: perturbs exterior paths to create intentional texture.

Use it when: texture is a design choice, improves grip, or masks small surface variations that are already understood.

Desired result: controlled, repeatable texture without dimensional loss at fits, holes, or first-layer edges.

- Scope chooses which surfaces receive texture.
- Mode chooses displacement, extrusion variation, or both.
- Noise type, feature scale, octaves, and persistence shape the texture.
- Point distance and thickness control frequency and amplitude.
- Ripple count, offset, and layers between offsets create organized variation.
- First-layer application extends texture to the bed-contact layer.

Do not use fuzzy skin as evidence that an underlying extrusion defect is fixed.

G-code Output

What: controls output verbosity, object labeling, cancellation metadata, filenames, and some retraction reduction.

Use it when: the printer, host, or post-processing workflow needs specific metadata or file organization.

Desired result: compatible, traceable G-code that preserves the controls required by the receiving system.

- `Reduce infill retraction` avoids some unnecessary internal retracts.
- Line numbers and comments aid certain workflows but increase file size.
- Object labels and exclude-object metadata enable supported per-object control.
- Filename format builds traceable names from placeholders.

Open a sample output and verify the actual header and commands before relying on metadata for automation.

Change Extrusion Role G-code

What: runs custom commands when the slicer changes feature role.

Use it when: an advanced, validated machine workflow genuinely requires a role transition command.

Desired result: deterministic commands at the intended transitions, with no unsupported firmware instructions or unsafe state changes.

Most filament tuning does not need this field. Treat pasted G-code as machine control software and review it line by line.

Post-processing Scripts

What: sends completed G-code through external programs.

Use it when: a documented workflow must transform, validate, sign, or route the output after slicing.

Desired result: reproducible transformed output, preserved safety commands, and a visible failure if the script cannot complete.

Scripts can silently change the file after preview. Archive both the source project and final sent G-code.

Notes

What: stores human-readable profile context.

Use it when: recording assumptions, calibration date, material lot, dependencies, or warnings that numbers alone cannot express.

Desired result: another operator can understand why the profile exists and what must be revalidated before reuse.

TinmanX1 Fiber and Strength Tools

Everything in this chapter is a **TinmanX1 addition** relative to OrcaSlicer 2.4.2. The settings are distributed across three profile types because they answer different questions:

- the **process profile** decides where and how fiber is planned;
- the **filament profile** describes the continuous fiber and carrier plastic;
- the **printer profile** describes FibreSeek hardware and machine commands.

Do not move a machine-contract value into process tuning. A part strategy may change from job to job; nozzle offsets, cutter travel, and tool-change commands must remain faithful to the hardware.

Strength Lens

What: provides a pre-slice color view of orientation risk using a simplified material model and chosen load axis.

Use it when: orienting a load-bearing part or deciding where continuous fiber could help. It is most useful for comparing orientations of the same design.

Desired result: critical loads run through strong printed roads instead of trying to separate the layer stack.

- `Strength Lens` enables the overlay.
- `Material model` selects filament-preset behavior, isotropic comparison, ordinary anisotropic FDM, or continuous fiber.
- `Load axis` selects Auto, X, Y, or Z.
- `Strength Lens payload` is saved analysis data. It is a **profile-contract field**, not a normal tuning control, and should not be hand-edited.

The lens is not finite-element analysis. It does not know the real boundary conditions, defects, voids, fiber waviness, temperature history, or measured material allowables. Use it to choose what to test, not to certify a part.

Process Profile: Planning

What: selects the manufacturing model and broad fiber layout for the part.

Use it when: choosing whether fiber supplements plastic roads or owns composite roads, and when establishing the first defensible layup before fine motion tuning.

Desired result: only intentional, manufacturable routes appear, beginning on the intended layers and aligned with the load strategy.

- `Fiber manufacturing mode` chooses **Plastic + fiber overlay** or **Fiber-owned composite roads**. Overlay preserves the polymer structure and adds fiber; composite-road mode lets fiber planning own selected roads and refill the remaining plastic regions.
- `Generate fiber perimeters` requests closed reinforcement near outlines.
- `Generate fiber infill` requests internal reinforcing routes.
- `Fiber reinforcement mode` applies the Light, Medium, or Heavy planning policy. Treat these as coordinated starting presets, not measured strength grades.
- `Fiber start layer` delays reinforcement until the part has a stable plastic foundation or until fiber reaches a useful Z region.
- `Fiber infill pattern` chooses the broad internal layout.
- `Fiber infill density` derives road spacing from line width when set above zero; zero delegates spacing to the mode or explicit spacing field.

- `Fiber infill angles` supplies a comma-separated angle sequence when the part needs known directions rather than the pattern default.
- `Fiber infill source` chooses plastic traces, newly generated ribs, or explicitly marked fiber paths.
- `Fiber seam position` chooses source, nearest, aligned, rear, or deterministic random start locations for closed routes.
- `Fiber seam angle` positions aligned starts around each loop.
- `Fiber print order` is an expert planner-order code. Leave it at the validated profile value unless a documented workflow requires a different sequence.

Choose route family and direction before density. More fiber in the wrong direction can add cost and complexity without solving the load case.

Process Profile: Geometry

What: constrains fiber-road width, spacing, layer schedule, curvature, minimum length, and clearance from walls.

Use it when: planned routes do not match the physical composite road, enter spaces the nozzle cannot reach, bend too tightly, or create impractically short segments.

Desired result: smooth, continuous routes that fit the machine's placement geometry and stay inside valid model regions.

- `Fiber line width` should match the effective deposited composite road, not merely the nominal fiber diameter.
- `Fiber infill spacing` sets explicit road pitch; zero delegates it to the reinforcement mode.
- `Fiber layer height` sets the fiber macro-layer spacing; zero uses the mode.
- `Fiber layer step` reinforces every Nth printable layer; zero uses the mode.
- `Fiber minimum radius` expresses the preferred bend quality. Larger values favor smoother paths, but the planner may still use shorter-radius paths when necessary.
- `Fiber max arc segment` controls how finely curves are divided. Short segments are smoother but produce more G-code and motion blocks.
- `Fiber minimum route length` filters generated routes; zero uses the planner default.
- `Fiber perimeter minimum length` specifically rejects short closed or outline-oriented routes.
- `Fiber hardware minimum length` protects the cutter/feed mechanism from segments shorter than it can manage reliably.
- `Fiber perimeter inset` places outline reinforcement away from the model boundary; zero delegates to the planner.
- `Fiber infill inset` keeps generated ribs away from walls and holes; zero delegates to the planner.

The mechanical minimum is a hardware constraint. Do not lower it simply to make more preview paths appear.

Process Profile: Feed and Motion

What: controls normal fiber speed, start behavior, tensioning, feed ratio, after-cut plastic, and optional Z lift.

Use it when: correct routes are visible in preview but physical fiber slips, buckles, detaches at starts, remains loose, or drags after a cut.

Desired result: the strand starts attached, stays placed without excess tension, and ends cleanly without disturbing the part.

- `Fiber print speed` is the normal deposition target; zero uses the planner default.
- `Fiber start speed` slows the beginning of a route; zero uses the default.

- Start, normal, and finish maximum speeds cap the three route phases.
- `Fiber start length` defines the initial controlled phase.
- `Fiber slow length` keeps the strand slow while adhesion develops.
- `Fiber tension length` defines the straightening segment.
- `Fiber tension feedrate` sets motion during tensioning.
- `Fiber tension release` returns a fraction of feed to reduce retained tension; zero disables release compensation.
- `Fiber feedrate` is fiber feed relative to path length. Below 100% adds tension; above 100% relaxes the strand. Small changes can have large physical effects.
- `After-cut plastic multiplier` adjusts polymer immediately after a fiber cut or restart.
- `Fiber Z hop after cut` lifts before travel to reduce dragging.

Do not compensate a wrong route length or cutter position with feed percentage. Fix geometry and hardware contracts first.

Process Profile: Advanced Feed Phases

What: places speed floors and curve-sensitive targets around start, normal, finish, and correction phases.

Use it when: a validated basic route still changes quality between straight and curved segments or trajectory corrections need dedicated motion behavior.

Desired result: smooth phase transitions with no stalls, sudden tension spikes, or correction marks.

- Start, normal, and finish `minimum speed` values are curved-segment targets.
- Their `speed floor` values are the lowest allowed targets.
- `Override correction move speed` activates dedicated correction motion.
- `Correction move speed` and `correction fiber feedrate` define that motion.

These are expert controls. Zero commonly delegates to the existing planner policy. Tune them only with synchronized video, G-code inspection, or other evidence that identifies the failing phase.

Process Profile: Fiber First Layer

What: supplies optional flow, width, height, and speed multipliers for fiber-related first-layer geometry.

Use it when: fiber itself is intentionally present near the first layer and the ordinary plastic first-layer settings do not produce a sound foundation.

Desired result: stable, dimensionally controlled first-layer fiber geometry without over-compression or poor bed clearance.

- `Fiber first-layer flow`, `width`, `height`, and `speed` are expert overrides. Zero delegates to the validated defaults.

Do not use these fields to correct the printer's global Z offset or bed mesh.

Process Profile: Planner Limits

What: caps route count and defines how fiber loops coexist with plastic shells and cutter cycles.

Use it when: a layup is too dense, produces excessive cut/restart events, or places fiber too close to the visible shell or internal voids.

Desired result: a bounded route plan with deliberate shell protection and a manageable number of cutter operations.

- `Fiber max routes per layer` is a safety cap; zero uses the reinforcement mode.
- `Fiber routes per cut` groups same-layer routes into cut/restart cycles where the hardware and path permit.
- `Fiber outer loops` requests reinforcement near outside-oriented contours.
- `Fiber inner loops` requests reinforcement near holes or inside contours.
- `Plastic loops outside fiber` retains polymer between fiber and the visible exterior.
- `Plastic loops inside fiber` retains polymer between fiber and infill or internal openings.

Preview every layer containing holes. An apparent loop count is not useful if the resulting bend radius or shell clearance is unmanufacturable.

Process Profile: Advanced Layup Payload

What: stores serialized planner instructions that cannot be expressed by the ordinary controls.

Use it when: an audited workflow generates explicit layup bands or solid fiber definitions programmatically.

Desired result: schema-valid, reproducible planning data that survives save, reload, and export without silent reinterpretation.

- `Fiber advanced layup payload` can encode layer/Z bands, mode, pattern, perimeter and infill intent, spacing, route caps, priority, and prime-line information.
- `Fiber solid infill payload` can encode solid-fiber angles, road width, extension, and minimum segment length.

Both are **profile-contract fields**. Most users should leave them unchanged. Hand-editing unvalidated JSON can create plausible-looking but unsafe output.

Filament Profile: Continuous Fiber

What: identifies the fiber spool and carrier plastic, then records material temperatures, speeds, timing, and cost data.

Use it when: creating or auditing a TinmanX1 composite filament preset for a specific fiber and polymer combination.

Desired result: one preset describes the actual installed materials and supports traceable estimates and machine commands.

- `Composite enabled` marks the preset as a composite workflow.
- Fiber name, type, manufacturer, diameter, linear density, spool length, and cost identify and quantify the reinforcement.
- Carrier-plastic name, type, manufacturer, diameter, density, cost, and spool weight identify the matrix material.
- Fiber-nozzle preheat and standby temperatures define the composite thermal state around use and idle periods.
- `Fiber first layers height` records the plastic foundation height before fiber placement.
- Plastic and fiber extrusion speeds describe material-specific feed behavior.
- `Fiber restart pause` gives the mechanism time to establish a restarted strand.
- `Fiber finish ironing distance` controls the end-of-route finishing distance.
- `Fiber priming line height` describes priming geometry.
- `Fiber material kind` classifies the reinforcement for planning and Strength Lens behavior.
- `Fiber source material ID` links the preset to a source material record.

Material identity is evidence, not decoration. Record lot, conditioning, and source ID so a good layup can be reproduced.

Printer Profile: FibreSeek Machine Contract

What: declares that a printer supports fiber, identifies the nozzle arrangement, and selects the post-processing contract.

Use it when: commissioning or auditing a FibreSeek-capable printer profile.

Desired result: the slicer describes the real machine exactly and emits only operations the machine can perform.

- `Fiber enabled` activates FibreSeek capability for the printer profile.
- `Shared fiber nozzle` states whether plastic and fiber share the same nozzle path.
- Plastic and composite nozzle diameters define the two road systems.
- `Fiber postprocessor type` selects the expected output transformation.

These are **profile-contract fields**. They should not vary per cosmetic print.

Printer Profile: Extruder Contract

What: records plastic/composite tool offsets, heat-up rates, and firmware-addressable fans.

Use it when: calibrating tool alignment, time estimates, or tool-specific cooling hardware.

Desired result: plastic and composite paths align in X, Y, and Z, thermal planning is credible, and the correct fans are addressed.

- Plastic and composite X/Y/Z offsets locate each extrusion path.
- Plastic and composite heat-up speeds support timing and preheat planning.
- `Has fan` and fan index identify each tool's controllable cooling output.

An incorrect offset is a collision and placement risk. Measure and validate it with dedicated registration tests.

Printer Profile: Thermal and Timing

What: records bed and chamber heat-up rates plus the firmware motion-buffer capacity used in planning and estimates.

Use it when: commissioning the machine or correcting consistently wrong thermal/time estimates.

Desired result: estimates and staged operations match observed machine behavior without overrunning the controller's intended motion capacity.

- Bed and chamber heat-up speeds are measured rates.
- Motion-block buffer size is a firmware capability value.

These are **profile-contract fields**, not speed-tuning shortcuts.

Printer Profile: Cut and Contact

What: defines cutter travel, restart feed, nozzle clearance, lane roles, and the commands surrounding cutting and fiber tool changes.

Use it when: commissioning, repairing, or formally updating FibreSeek hardware behavior.

Desired result: reliable cut/restart cycles, collision-free placement, and correct state transitions.

- `Fiber cut distance` must match the cutter mechanism.
- `Fiber restart length` restores fiber after a cut.
- Normal and extended nozzle contact radii reserve clearance around the physical placement tool.
- `Fiber slot roles` maps lanes to plastic, composite, or fiber roles.
- Cut G-code and before/after-toolchange G-code are machine commands.
- `Fiber machine contract` is serialized capability data.

Every field in this group is a **profile-contract field**. Review custom G-code line by line and prove changes on a safe commissioning artifact.

Printer Profile: Continuous Fiber Lane

What: identifies the reinforcement loaded in the machine lane independently of the process strategy.

Use it when: associating a physical lane with a material record.

Desired result: the machine lane, filament preset, estimates, and exported metadata all refer to the same fiber.

- Name, type, material kind, source material ID, diameter, and linear density define the lane contents.

TinmanX1 Wave Overhangs

Wave Overhangs are another TinmanX1-only strength/support tool, but they are not a fiber feature. Their complete group-by-group guide is in the Support tab. Keep the two systems conceptually separate: wave paths try to support geometry during printing; continuous fiber routes try to reinforce the finished part.

Fiber Validation Gate

Before releasing a fiber profile:

1. Confirm the machine contract and tool offsets against the physical printer.
2. Confirm material identity and conditioning.
3. Preview every reinforced layer, route start, cut, restart, hole, and narrow region.
4. Verify minimum radius, minimum route length, and contact clearance.
5. Print a disposable placement coupon before a valuable part.
6. Section or destructively test representative samples when strength matters.
7. Save the project, final G-code, profile revisions, and exact TinmanX1 commit.

Motion Diagnostics: Acceleration, Input Shaping, and VFA

Ringing, frame motion, vertical fine artifacts, and geometry showing through a wall can all look like vertical bands in a photograph. They do not respond to the same setting. Use the shape, direction, and repeatability of the evidence to choose the test before changing the profile.

Separate the Failure Families

Evidence	Best first test	What the test can establish	What it cannot establish
Ripples trail a sharp corner	ringing tower or input-shaper measurement	dominant resonance and residual corner echo	best wall speed or melt limit
A layer shifts and every feature above it moves	conservative acceleration/step-loss test and mechanical inspection	loss of registration or mechanical instability	cosmetic optimum below the failure point
Fine vertical texture changes with wall speed	VFA speed sweep	quieter and louder speed ranges	maximum volumetric flow by itself
Bands align with hidden ribs, holes, or infill	representative wall coupon	internal-feature print-through	general machine resonance
Gloss changes where feature roles change	sliced-toolpath review	speed, flow, cooling, or acceleration transition	axis calibration error

Do not call every repeated mark ringing. Ringing follows a disturbance and decays away from it. VFA often continues along a nominally steady wall. Hidden geometry print-through remains registered to the feature behind the wall.

Input Shaping Is Not an Acceleration Limit

An accelerometer-based shaper calibration identifies resonant frequencies and estimates the smoothing introduced by candidate filters. The reported "recommended maximum acceleration" is a smoothing guideline for that shaper, not proof that the printer will hold registration, produce its best surface, or extrude correctly at that value.

Use measured input shaping to reduce a known resonance. Then validate motion with a printed part. Save the axis, shaper type, frequency, damping value when available, accelerometer mounting, toolhead configuration, and date. Re-run the measurement after changing toolhead mass, belt path, frame stiffness, or major gantry components.

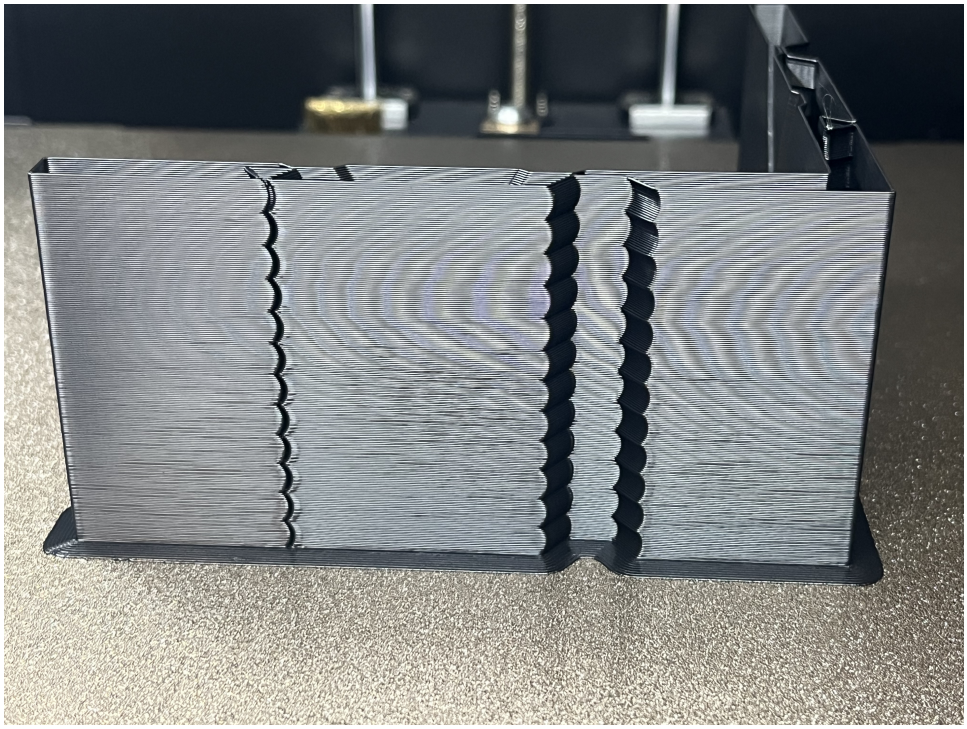
Run an Acceleration Tower as a Controlled Experiment

1. Use a model with deliberate sharp features on both axes and enough height for several clearly labeled bands.
2. Hold wall speed, layer height, line width, temperature, flow, PA, cooling, wall order, and material condition fixed.
3. Start below the current production acceleration and stop below known motor, frame, firmware, and shaper limits.
4. Confirm in the G-code or firmware log that the intended acceleration owns each band. A later slicer `M204` or `M205` command can silently override an earlier firmware tuning command. Audit the last effective writer, then verify live state after extrusion begins.
5. Compare broad faces, corner echoes, registration, and dimensional shape.

The first band with skipped steps, layer displacement, or growing skew is a mechanical failure boundary. Apply margin below it. A stable highest band is not automatically the best profile value.

How to read a tower with no visible knee

If every band remains registered and the corner echoes do not change clearly, record the tested range as mechanically stable and stop. Do not select the top number merely because it printed. The test did not resolve a cosmetic optimum. Use a geometry-representative comparison or change test families.



Acceleration tower viewed from the deliberate scallop and corner features. The repeated echoes remain similar through the height, so the tower establishes mechanical stability but not a cosmetic acceleration knee.

Use a VFA Sweep for Speed-Dependent Texture

A vertical-fine-artifact test changes outer-wall speed in height bands while printing repeated curves and angled faces. It is useful when broad curved walls show vertical texture but axis-aligned faces or an acceleration tower remain comparatively smooth.

Before the test:

- dry and identify the filament;
- settle temperature, maximum volumetric flow, PA, and global flow;
- use the production nozzle, layer height, line width, cooling, and input shaping;
- choose one fixed acceleration that is already mechanically credible;
- disable adaptive speed features that would blur the commanded bands;
- keep the model orientation fixed for every comparison.

Choose the upper speed with the volumetric-flow check:

```
required flow = line width x layer height x speed
```

For a `0.60 mm` line, `0.30 mm` layer, and `80 mm/s` wall, the nominal request is `14.4 mm3/s` before slicer-specific width and flow adjustments. The test is not valid if the slicer caps the upper bands at the filament's maximum volumetric flow. Inspect the preview or emitted feed rates.

Reading the VFA tower

1. Mark the speed represented by every height band.
2. View each major face under diffuse light and then glancing light.
3. Look for texture that begins, ends, or changes wavelength at a speed boundary.
4. Prefer the middle of a broad quiet region over a single unusually clean line.
5. Reject a visually quiet band if it approaches the melt-flow ceiling, weakens layer bonding, or damages dimensions.
6. Confirm the candidate on the original geometry.

If every band looks alike, the result is still useful: steady wall speed is not the dominant variable in the tested range. Investigate internal-feature print-through, wall thickness/order, belt or roller periodicity, and feature-role transitions before changing material flow.

Use wall angle to localize a CoreXY motion source

A multi-vane VFA specimen can do more than select a quiet speed. On a CoreXY machine, different wall headings load the two motor-and-belt loops by different amounts. If the same height band is quiet on one heading and noisy on another, the contrast can separate a motion path from a filament-wide problem.

For the common ideal CoreXY transform, a unit move at heading `theta` gives relative loop demand:

```
A = cos(theta) + sin(theta)
B = cos(theta) - sin(theta)
```

Use the absolute values when comparing demand. Motor names, signs, and which physical belt is called A or B vary by machine, so verify the printer's routing before naming a component. Also derive the actual line heading from the model or G-code; an embossed vane label can describe the panel arrangement rather than the emitted segment direction.

Actual wall heading	Relative loop A	Relative loop B	Diagnostic value
0 degrees	1.000	1.000	both loops equally
30 degrees	1.366	0.366	A dominant, not exclusive
45 degrees	1.414	0.000	A isolated in the ideal transform
90 degrees	1.000	1.000	both loops equally, with a different Cartesian direction
135 degrees	0.000	1.414	B isolated in the ideal transform

This comparison is strongest when the specimen includes both `45` and `135` degree walls. Without the opposite diagonal, one loop can be isolated while the other remains only dominant, which limits the conclusion.

Classify the mark before assigning it to a loop:

- broad periodic waves that continue through a constant-speed wall field are motion evidence;
- beads, pits, or curls confined to a seam, free edge, corner, or band change remain pressure-advance, retraction, cornering, and thin-wall candidates;
- progressive thinning or missing extrusion across every heading remains a melt-flow or feed-path candidate;
- a band fixed at the same Z height on every heading is more consistent with a layer event or Z-related source than an XY-loop order.

Do not stop at a visual angle comparison. Test whether the wavelength or onset speed matches a physical order. For belt pitch `p`, pulley tooth count `N`, and linear belt speed `v_belt`:

```
tooth-pass frequency = v_belt / p
one-pulley-revolution distance = N x p
```

On an axis-aligned CoreXY move, each active loop commonly runs at the Cartesian wall speed. On a single-loop `45` or `135` degree move, the active loop runs at approximately `sqrt(2)` times Cartesian speed. A belt-related frequency should therefore appear at about `1/sqrt(2)` of the axis-wall Cartesian speed on the single-loop diagonal. The corresponding wall-space orders are `p` versus `p/sqrt(2)` for one tooth and `N x p` versus `N x p/sqrt(2)` for one pulley revolution.

A matching order is evidence, not a verdict. A plucked-belt tuning frequency is not guaranteed to equal the loaded operating mode, and the same order can be amplified by pulley eccentricity, idler runout, belt-edge contact, gantry compliance, or rail preload. Preserve the current state before touching it, then inspect in this order:

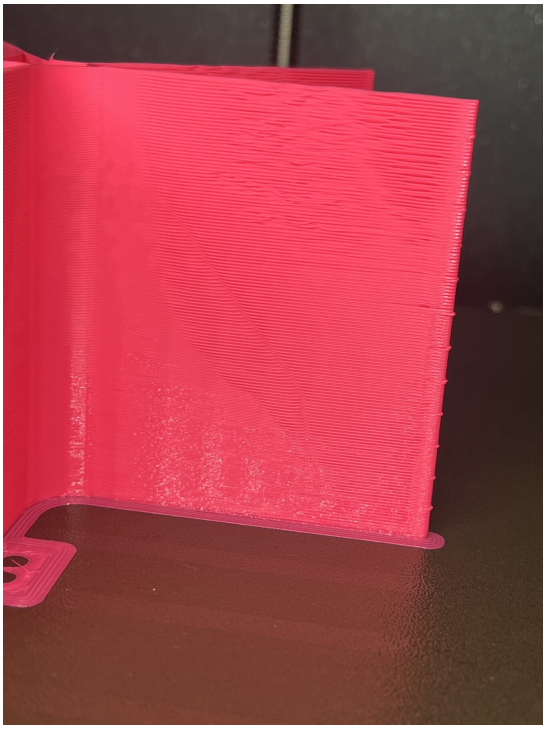
1. Both motor pulleys: set screw on the shaft flat, second screw tight, correct axial height, no axial walk, low runout, and full-revolution clearance.
2. Every idler and spacer in each loop: no notchiness, axial play, wobble, debris, tooth-on-smooth-idler contact, or persistent belt-edge witness mark.
3. Belt planes and anchors: no twist, edge polish, fray, vertical tracking change, asymmetric toolhead seating, or loose clamp.
4. Shared gantry interfaces: square geometry, X-rail mounting stress or tight spots, carriage rock, Y-guide drag, and position-dependent cable or PTFE load.

Do not repeatedly retension by feel between comparison prints. Record the before state, change one identified condition, re-run any motion compensation invalidated by that change, and repeat the same final G-code. A compact `0/45/90/135` degree control provides both equal-loop and isolated-loop views.

Bounded field finding: GT1.5 conversion

One controlled CORE One L-to-L+ investigation used GT1.5 belts, 21-tooth motor pulleys, and byte-identical `40-160 mm/s` VFA G-code across repeated prints. The walls remained fully fed after temperature, flow, pressure advance, and maximum volumetric flow had been settled. Gantry squaring, belt tuning, phase stepping, input shaping, homing, Z alignment, and load-cell checks did not remove the direction-dependent face waves.

The three photographs below are from the same completed job and the same final G-code. Read each wall from the lower `40 mm/s` band toward the upper `160 mm/s` band. Lighting and camera angle differ, so use them to compare onset and pattern family rather than to calculate a numerical amplitude ratio.

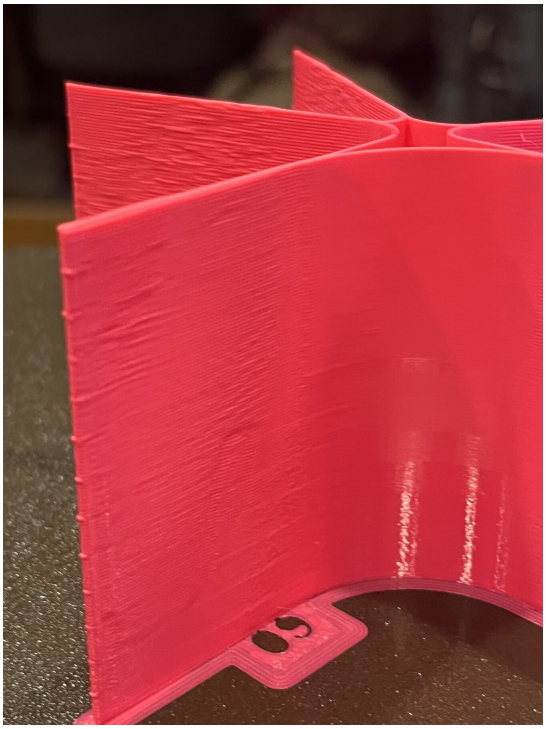


The 90 degree wall loads both CoreXY loops equally. Its lower bands are comparatively calm, while broad diagonal waves become prominent through the upper, faster bands.

Axis-aligned walls were worst in the `130-160 mm/s` bands. With `1.50 mm` pitch, those speeds produce `86.7-106.7 Hz` tooth pass; the `130` and `140 mm/s` bands produce `86.7` and `93.3 Hz`, inside the machine's documented `85-95 Hz` belt-tuning range. The isolated `45` degree face began showing the same family around `90-100 mm/s`, close to the `90.2-100.8 mm/s` range predicted by the `sqrt(2)` loop-speed shift.



The actual 45 degree wall isolates one loop in the ideal CoreXY transform. The broad packets begin at a lower Cartesian speed than on the equal-loop axis wall, which is the diagnostic shift predicted by the square-root-of-two loop-speed relationship.



This vane is labeled 60 degrees by the source layout, but its emitted wall segment is approximately 120 degrees and is dominated by the opposite loop. It also carries the high-speed wave family, so the evidence does not support only one affected loop.

That agreement promoted the GT1.5 pulley, idler, belt-plane, and shared-gantry interfaces above more filament tuning. It did **not** prove that a particular pulley or belt was defective, and the investigation had not yet completed the one-component correction test at publication time. This is the appropriate claim boundary for a frequency-and-angle correlation.

Worked Field Example

A moving-bed printer produced dimensionally correct ABS parts but showed broad vertical texture on a large curved cover. Its production outer wall used `48 mm/s` and `1200 mm/s2`. Measured input shaping was active. A conservative `300-1500 mm/s2` acceleration tower retained registration through every band, and the deliberate corner echoes did not develop a repeatable step change.

The correct conclusion was limited: the machine was mechanically stable through the tested acceleration range under those conditions. The tower did not prove that `1500 mm/s2` was ideal, and it did not support changing the already validated flow or dimensional compensation. The next test held acceleration at `1200 mm/s2` and swept outer-wall speed from `20` to `80 mm/s` in seven 5 mm bands.

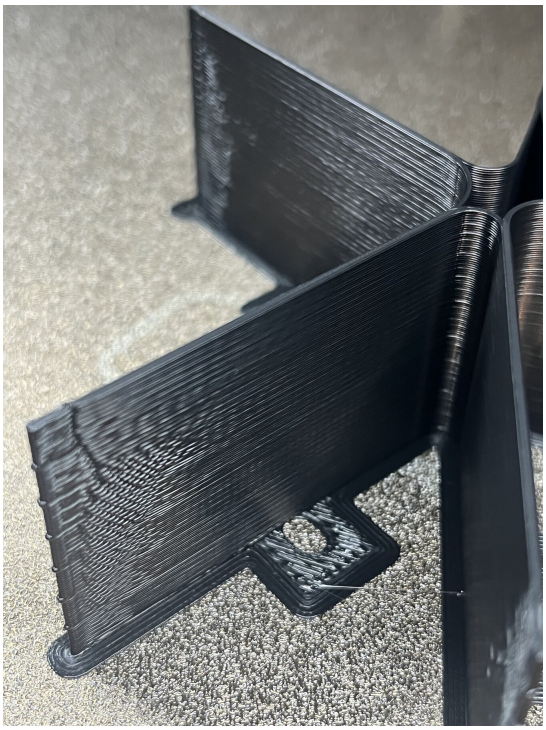
That follow-up also exposed a useful preflight lesson. Its machine-contract block requested square-corner velocity `1 mm/s`, but a later slicer `M205` line became the effective writer and live firmware reported `10 mm/s`. The test remained controlled because the value stayed fixed across every speed band, but the record had to use the live value. Do not infer active motion state from the first matching command in a file.

Physical VFA result

The printed sweep resolved a useful quality window. Read this particular artifact by height: the final file changed every outer wall from `20` to `80 mm/s` in 5 mm bands from bottom to top. The embossed speed numbers around the base came from the source model and did not identify the active speed of each vane after postprocessing.

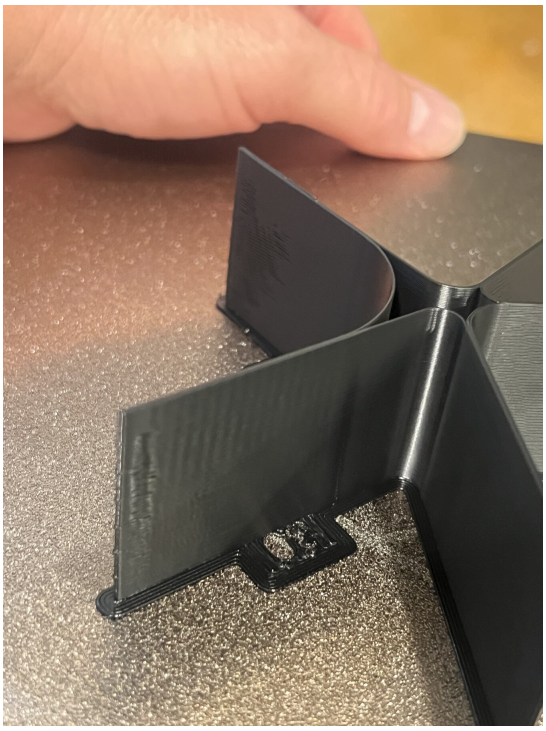
Under both diffuse and glancing light, the coarsest repeating texture appeared from `30-50 mm/s`. The `60 mm/s` band improved, while `70-80 mm/s` formed the broadest quiet region. The result was also direction-sensitive: some vane

orientations displayed the pattern much more strongly than others at the same height. That combination supports a speed-dependent motion interaction rather than random moisture, global flow error, or Z-axis banding.



The most revealing vane orientation shows coarse texture in the lower and middle speed bands, followed by a quieter upper region. Read the bands from bottom to top, not from the embossed base labels.

The selected candidate was `70 mm/s`, with acceleration held at `1200 mm/s2`. Although `80 mm/s` was often visually competitive, its nominal flow request was `14.4 mm3/s` against a recorded `15 mm3/s` material limit. Choosing `70 mm/s` preserved more melt-flow margin and followed the rule of using the middle of a broad quiet region.



A differently oriented vane is quieter through much of the same height, showing why every major face must be inspected before choosing a VFA band.

Do not score the unsupported free-edge loops as broad-wall VFA. The thin test edge and abrupt speed changes can expose pressure and corner-velocity transients that are not representative of a closed production wall. Judge the field away from the edge, then validate the selected speed on the real geometry.

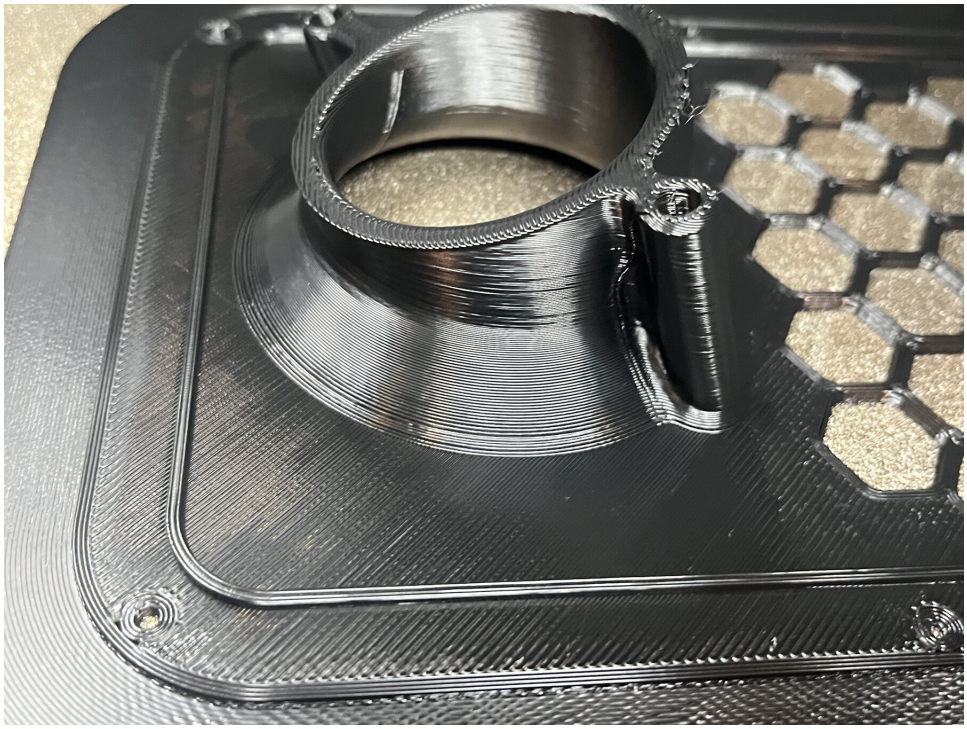
Representative-part confirmation

The same machine then printed the original curved cover with only the normal outer-wall speed changed from `48` to `70 mm/s`. First-layer walls retained their original speed, acceleration remained `1200 mm/s2`, pressure advance remained `0.025`, and the production nozzle, layer height, line width, temperature, cooling schedule, wall order, flow, and input shaping were held constant. The uploaded and downloaded G-code matched byte-for-byte, and live firmware state confirmed the intended contract during the print.



The completed production cover on the printer bed. This representative geometry, rather than the calibration tower alone, confirmed the selected wall-speed range.

The broad repeating texture on the production curves was substantially reduced. That confirmed `70 mm/s` as the production outer-wall value for this exact machine, material, `0.6 mm` nozzle, `0.30 mm` layer, and `0.60 mm` line-width profile. It is not a general maximum or a universal FibreSeek setting.



The curved collar is smooth through most of its height after the 70 mm/s representative validation, confirming that the broad VFA was speed-dependent.

One horizontal texture and gloss band remained on a tall wall. Toolpath review showed no change to that wall's XY geometry or commanded outer-wall speed at the band. Other regions on the same layers changed from wall-dominated motion to solid infill, bridge, and top-surface work; layer time fell and normal fan commands increased. That makes cooling and layer context the stronger next hypothesis.



A localized horizontal band remains even though the broad curved-wall VFA improved. Treat this as a layer-context or cooling transition, not proof that the XY curve needs finer Z layers.

Adaptive layer height is not the right first tool for this vertical curve. The curve is traced in XY on every layer, so changing Z height does not add XY facets or correct a speed-dependent wall texture. Adaptive layers help sloped or curved surfaces whose shape changes with Z. For this residual band, keep geometry and motion fixed and run a one-variable cooling comparison instead. The next validation therefore holds ordinary part cooling at 20% while preserving the original 100% bridge-fan pulses and fan-off commands.

This example is evidence about one machine, material, nozzle, and setup. Its numbers are not universal recommendations. The reusable lesson is the decision logic: a completed high band proves survival, while a visible and repeatable knee is needed to select a cosmetic limit.

OrcaSlicer and TinmanX1 Version Note

The TinmanX1 `2.4.2` build documented in this edition provides a VFA dialog with start speed, end speed, and step size. Its built-in test uses a fixed 35 mm model and changes speed every 5 mm. It does not expose the newer nozzle-aware auto-scaling options described for OrcaSlicer builds after `2.4.2`. With a larger nozzle or layer height, verify model size, requested volumetric flow, actual feed rates, and the number of usable bands before printing.

Do not assume a later OrcaSlicer calibration screen and TinmanX1 `2.4.2` create identical geometry or ranges. Record the application version with every saved test. See the version basis and the Speed tab for the controls that remain active around the calibration.

Minimum Experiment Record

Keep these fields with every motion test:

- printer kinematics, tool, nozzle, and build-plate orientation;
- slicer name/version and firmware version;
- source model and final G-code hashes;
- material, drying state, temperature, cooling, flow, PA, and MVS;
- input-shaper settings and square-corner or jerk setting;
- fixed speed or acceleration and the exact band map;
- photographs of at least two faces in consistent lighting;
- accepted range, rejected range, uncertainty, and next isolated test.

Profile Integrity and Experimental Resets

A controlled test is only controlled when the printer, tool, nozzle, filament, and process geometry are the ones the experiment claims to use. A slicer can display a familiar profile name while inherited values, a stale project, firmware state, or the selected physical tool describe a different setup.

This chapter explains how to recognize that failure, preserve the useful evidence, and reset without discarding the whole tuning history.

The Four Identities That Must Agree

Before accepting a tuning result, verify four independent descriptions of the job:

Identity	What to verify
Physical machine	Installed tool, actual nozzle diameter, nozzle condition, filament path, and active material
Printer profile	Compatible machine, tool index, nozzle diameter, firmware flavor, limits, offsets, and start G-code
Filament profile	Machine compatibility, temperature, flow, PA, MVS, cooling, retraction, and compensation
Process and output	Layer height, line widths, wall order, seam controls, selected tool, and the commands emitted in final G-code

The profile label is evidence of intent, not proof of execution. Confirm the resolved settings and the emitted file. For networked printers, compare the important commands with live firmware state when practical.

Preflight Identity Gate

Run this gate before the first calibration and after any nozzle, hotend, firmware, slicer, or profile change:

1. Physically identify the selected tool and measure or verify its nozzle.
2. Confirm that the printer profile assigns that diameter to the same tool.
3. Confirm that layer height and every important line width are credible for that nozzle.
4. Confirm that the filament profile is compatibility-gated to the intended machine, tool, and nozzle.
5. Slice a small known model and inspect the final G-code for tool selection, temperatures, PA, MVS, fan behavior, line geometry, and unexpected material or auxiliary-tool commands.
6. Start a fresh project or restart the slicer if cached project state does not agree with the saved profiles.

Do not begin fine tuning until all six checks agree.

Isolate Concurrent Slicer Sessions

When several printers are being tuned at once, do not reuse one live slicer process for more than one physical print. Open a distinct application instance with its own project and state for each job, and name the output after the machine, material, tool, and purpose. A fresh project inside a shared process is not always sufficient: cached tool assignments, inherited materials, and queued uploads can survive between projects.

Before launching each physical print, home the printer and verify that the remote job is actually associated with the intended uploaded file. An upload or start request that times out is not proof of failure or success; check the printer's live filename, state, tool, and targets before proceeding.

Audit the Whole Tool Lifecycle

A multi-tool job is not safe to promote merely because its first tool command is correct. Inspect the complete executable G-code, including start macros, object assignments, transition/tool-change macros, purge or cutter routines, and end G-code. Verify that each command which selects, loads, unloads, cuts, or purges filament maps to the intended physical path.

This matters especially when a profile is remapped from a stale tool or a different material system. An otherwise successful specimen can end by trying to unload from an unintended lane, leaving the printer paused or faulted after the geometry is complete. Treat that as a workflow defect to fix before the next job, not as evidence that the selected material path worked.

Warning Signs of a False Baseline

Stop and re-audit identity when:

- many seam settings move or spread the defect but never remove it;
- a modest setting change creates a disproportionately large visual change;
- the best value keeps landing at the edge of the tested range;
- line widths or layer heights look appropriate for a different nozzle;
- a project behaves differently after reopening even though the named profile appears unchanged;
- final G-code selects a different tool or emits commands not expected for the intended material path;
- a formerly stable profile changes immediately after hardware or firmware work;
- machine faults, failed probing, or nozzle damage make the baseline non-repeatable.

Repeated machine faults are not filament data. Suspend material tuning until the machine can repeat its baseline, and record those events separately from accepted and rejected material experiments.

Worked FibreSeek Seam Reset

A FibreSeek ABS tuning sequence produced a broad disturbed band at the aligned rear seam of a circular part. Conventional seam gaps, scarf height, scarf length, scarf step count, scarf flow, scarf speed, entire-loop scarfing, inner-wall scarfing, and staggered inner seams were tested one variable at a time.

The trials taught useful local lessons:

- scarfing moved or spread the disturbance but did not remove it;
- more scarf steps made the transition worse;
- reducing scarf flow past the best trial reopened the defect;
- longer and entire-loop scarfing distributed the error over more wall;
- staggered inner seams created several disturbed regions instead of one;
- increasing conventional seam gap eventually opened the seam while retaining adjacent ridges.

Those are valid observations about the failure mode, but they were not valid profile-acceptance results. A later audit found that the coupons had been sliced as if the plastic tool used a `0.6 mm` nozzle, with `0.30 mm` layers and approximately `0.60/0.63 mm` outer/inner line widths. The installed plastic tool and the machine configuration identified that nozzle as `0.4 mm`.



A rejected FibreSeek seam trial made under the mismatched 0.6 mm process assumption. Staggering distributed the disturbed transition across a broad wall region. This photograph is diagnostic evidence, not an accepted profile result.

The experiment was reset on the same geometry with the true `0.4 mm` plastic tool, `0.20 mm` layers, `0.40/0.42 mm` outer/inner line widths, the established `0.030` pressure-advance value, inner-before-outer walls, an aligned rear seam, and no scarf, stagger, or wipe. The broad disturbed zone collapsed into one narrow, straight, repeatable seam while the surrounding wall remained smooth.



The same FibreSeek geometry after restoring the correct 0.4 mm machine and process baseline. The remaining seam is narrow and repeatable, making ordinary seam-gap refinement meaningful again.

The improvement came from restoring the correct baseline, not from discovering a universal FibreSeek seam value. The next seam-gap adjustment remained a candidate until a functional production part could confirm it.

How to Reset Without Losing the Experiment

1. Freeze the current files and mark their status honestly.
2. Label results made under a mismatched identity as `rejected - invalid baseline`, not merely `bad print`.
3. Preserve observations that remain useful, such as whether scarfing spread a disturbance or whether a fan change had no effect.
4. Restore the last verified physical and profile identity.
5. Re-slice the exact same geometry with conservative, ordinary settings.
6. Change no cosmetic control until the corrected baseline prints repeatably.
7. Resume with the smallest one-variable test that addresses the remaining defect.

This procedure separates two questions: whether a setting can influence the symptom, and whether its numerical value is valid for the intended setup.

Keep Tuned Profiles Machine-Specific

A filament profile contains machine-dependent values even when the spool is the same. PA, MVS, retraction, cooling response, temperature under flow, and dimensional compensation depend on the extruder, hotend, nozzle, duct, motion, and enclosure.

Use a distinct profile for each validated machine/tool/nozzle combination:

```
Material + printer + tool/nozzle + purpose + revision
```

Compatibility-gate the profile to that printer definition. Preserve the generic or supplier profile as a parent or untouched reference. Do not let a tuned FibreSeek profile silently appear as a valid choice for another printer.

Promotion Gate

Before calling a reset profile accepted:

- verify the selected physical tool and nozzle again;
- inspect resolved printer, filament, and process values;
- inspect final G-code for effective tool, PA, temperatures, fan commands, line geometry, and unintended auxiliary-tool commands;
- reproduce the corrected result from a clean project;
- print a representative functional part;
- measure required dimensions and fits after cooling;
- record provisional values separately from accepted values.

The functional part is the final judge. A calibration ring can show that the seam transition is controlled, but it cannot prove fit, strength, long-layer behavior, or production reliability.

Diagnostic Matrix

Use this table to select the first controlled experiment. It does not replace inspection of the machine baseline.

Symptom	First suspects	Check next	Avoid changing first
Open top channels, clean walls	top flow, top width, top speed	shell thickness, infill support	axis scale
Rough ridged top everywhere	global/top flow high	PA, first-layer carry-through	shrinkage
Thin or missing lines at high speed	MVS, temperature, partial clog	extruder grip, spool drag	dimensional compensation
Consistent under-extrusion at all speeds	flow ratio, diameter, obstruction	rotation distance, nozzle	cooling alone
Bulged non-seam corners	PA low	acceleration, wall order	global flow alone
Starved approach to corners	PA high	flow transitions, MVS	retraction alone
One vertical scar	seam transition	PA, retraction, wipe, gap	whole-model flow
Seam changes move or spread a broad defect	tool/nozzle/profile mismatch	resolved widths, layer height, final G-code	more scarf or stagger variables
Several scars after staggering	one transition error distributed across walls	aligned baseline, nozzle identity, PA	randomizing again
Ringing after corners	input shaping, belts, rigidity	acceleration	flow ratio
Random pits or bubbles	moisture, contamination	temperature, nozzle debris	seam placement
Fine hairs everywhere	moisture, temperature	retraction, travel	dimensions
Thick strings between features	retraction, travel time, ooze	temperature, wipe	cooling alone
Gap after travel	retraction too high, restart	PA, clog, spool drag	more retraction
Weak layers	temperature, moisture, excess cooling	production flow, chamber	seam settings
ABS/ASA warp or split	chamber, drafts, bed adhesion	fan, geometry, heat soak	axis steps
Poor bridge with good walls	bridge flow/speed/fan	bridge PA, line width	global flow first
Curling overhang edge	cooling, speed, temperature	wall order	shrinkage
Uniform proportional XY error	material shrinkage	larger artifact, both axes	contour offset
Same external offset at many sizes	contour compensation	PA and flow	material scale
Undersized holes only	hole compensation, polyholes	seam, orientation	global XY scale
Bottom layers too wide	elephant foot, Z, bed heat	initial-layer flow	whole-part scale
X and Y differ	mechanics, squareness, airflow	axis motion, orientation	one shared shrink value
Z error grows with height	Z motion or Z material scale	thermal/mechanical repeatability	XY compensation
Only first layer fails	Z offset, bed preparation	plate temperature, mesh	global flow
Gloss band at speed change	flow/temperature/cooling transition	MVS, acceleration	random seam
Repeating horizontal bands	Z mechanics, thermal cycles	spool drag, layer timing	hole compensation
Extruder clicks	MVS, temperature, clog, path drag	tension, nozzle, spool	PA cosmetic tuning
Good small print, bad large print	chamber, warp, long-flow demand	heat soak, MVS	retraction tower
Good calibration, bad real part	test not representative	actual speeds, widths, cooling	arbitrary multiplier

Triage Order

When several symptoms appear together:

1. Stop for unsafe motion, thermal faults, collisions, or severe adhesion loss.
2. Check filament dryness and nozzle condition.
3. Confirm the physical tool/nozzle, machine baseline, resolved profiles, and final G-code agree.
4. Determine whether the defect is global, feature-specific, speed-dependent, directional, layer-dependent, or localized to a seam.
5. Run the smallest test that isolates the leading cause.
6. Preserve the rejected result and change only one causal variable.

Useful Classifications

- **Global:** appears on every face and feature.
- **Directional:** depends on X/Y face or fan direction.
- **Rate-dependent:** begins at higher speed or flow.
- **Thermal:** changes with layer time, chamber state, or height.
- **Transition-localized:** appears at corners, starts, stops, or seams.
- **Bottom-only:** first-layer/Z/bed interaction.
- **Geometry-specific:** holes, bridges, overhangs, or top surfaces only.
- **Identity-dependent:** appears after a tool, nozzle, firmware, slicer, or profile change and resists ordinary one-variable tuning.

This classification often identifies the correct settings family before a single value is changed.

Glossary

Acceleration Rate at which toolhead speed changes, usually in `mm/s^2`. It affects pressure transients, cornering, and whether a short move reaches its requested speed.

Elephant foot Enlargement of the lowest layers caused by Z compression, heat, or first-layer flow. Correct it locally rather than scaling the whole part.

Extrusion multiplier / flow ratio A multiplier on commanded material. `1.00` is `100%`; `0.98` is `98%`.

FFF / FDM Layer-by-layer material-extrusion printing. FDM is a common industry term and a Stratasys trademark; FFF is the generic term used by many open-source projects.

Filament shrinkage compensation A per-material model scale intended to offset proportional dimensional change after cooling.

Heat soak Time or temperature condition used to let the bed, chamber, frame, and tool reach a repeatable thermal state before calibration or printing.

Line width The nominal width of a deposited path. It influences required flow, geometry, strength, and speed.

Maximum volumetric speed (MVS) The profile limit for requested plastic volume per second, in `mm^3/s`.

Pressure Advance (PA) / Linear Advance Firmware compensation for lag in nozzle pressure during speed changes.

Retraction Temporary reverse or relief motion in the extrusion path during non-print travel, intended to reduce ooze.

Seam The start/stop transition where a perimeter closes or a new layer begins.

Top-surface flow A feature-specific multiplier applied to top solid infill, usually multiplied by the global filament flow ratio.

Volumetric flow Requested deposited volume per second:

```
Q = speed * line width * layer height
```

XY contour compensation A near-constant offset applied to external contours.

XY hole compensation A near-constant offset applied to internal holes or contours.

Formula Reference

```
volumetric flow = speed * line width * layer height
```

```
maximum speed = MVS / (line width * layer height)
```

```
measured percentage = measured dimension / nominal dimension * 100
```

```
effective correction scale = 100 / measured percentage
```

```
effective feature flow = global flow ratio * feature flow ratio
```

Filament Tuning Experiment

Identity

Status: candidate | accepted | rejected | superseded
Experiment ID:
Date and time:
Operator:
Goal:

Hardware and Environment

Printer / firmware:
Tool / nozzle diameter / nozzle material / wear state:
Extruder / hotend:
Build plate / cleaning / adhesive:
Bed mesh / Z offset / tool offset:
Chamber state / soak / ambient:

Filament

Manufacturer / product / material / color:
Diameter / lot / date opened:
Drying cycle:
Storage and feed path:

Software and Artifacts

Slicer and version:
Printer profile revision:
Filament profile revision:
Process profile revision:
Project path or URL:
G-code path or URL:
G-code checksum:
Photo paths or URLs:
Photo orientation and lighting:

Controlled Test

Fixed variables:
Changed variable:
Values or range:
Expected observation:
Acceptance criterion:

Results

Raw observations:
Raw measurements and instrument:
Calculation:
Uncertainty or limitations:
Unexpected events:

Decision

Decision and reason:
Values promoted:
Values still provisional:
Next test:
Reviewer:

Filament Profile Release Checklist

Traceability

- [ ] Filament identity, color, lot, and condition are recorded.
- [ ] Printer, tool, nozzle, firmware, and slicer versions are recorded.
- [ ] The installed physical tool and nozzle match the printer definition.
- [ ] Resolved layer heights and line widths are credible for that nozzle.
- [ ] The filament profile is compatibility-gated to this machine/tool/nozzle.
- [ ] Parent printer and process profiles are identified.
- [ ] Every promoted value links to an accepted experiment.
- [ ] Project files, final G-code, photos, and measurements are preserved.
- [ ] Profile purpose and known limitations are stated.
- [ ] Each physical print used an isolated slicer session/project, with no concurrent-printer state carried into the job.

Calibration Order

- [ ] Machine baseline and first layer are repeatable.
- [ ] Filament was dried or its known-dry condition was documented.
- [ ] Temperature optimum was bracketed, not selected at an untested endpoint.
- [ ] Maximum volumetric flow includes a suitable production margin.
- [ ] Pressure/Linear Advance was tuned at representative conditions.
- [ ] Global flow produces continuous surfaces without ridges.
- [ ] Feature-specific flow changes are justified separately.
- [ ] Cooling and bridge settings preserve required adhesion.
- [ ] Retraction is the shortest reliable value for the tested path.
- [ ] Dimensions were tuned only after extrusion stabilized.
- [ ] Seam settings were tuned after PA, flow, and retraction.

Validation

- [ ] Broad top surface passes.
- [ ] External X/Y/Z dimensions pass after cooling.
- [ ] Internal holes and required fits pass.
- [ ] Elephant foot is within the defined limit.
- [ ] Bridges, overhangs, and small features pass.
- [ ] Representative production part passes.
- [ ] Mechanical comparison passes where strength matters.
- [ ] G-code review confirms the expected tool, temperatures, fans, PA, flow, MVS, mesh, offsets, and compensation.
- [ ] G-code contains no unintended tool-change, fiber, auxiliary, or calibration commands.
- [ ] Start, transition, and end G-code map every load, purge, cutter, and unload operation to the intended physical tool path.
- [ ] The printer was homed before launch and live state confirms the intended uploaded file is executing.
- [ ] Important emitted commands agree with effective live firmware state where that state is available.
- [ ] A clean project reproduces the result without calibration overrides.

Release

Profile name:
Version:
Optimization target:
Release date:
Author / reviewer:
Accepted operating envelope:
Known limitations:
Supersedes:

Sources and Further Reading

This guide favors official documentation and well-documented experimental work. Pages can change over time, so record the access date when using a source for a formal experiment or publication.

Last source review: 2026-10-08.

Prusa Research

- Extrusion multiplier calibration
- Drying filament
- Warping
- ABS material guide
- Under-extrusion
- Calibration index
- CORE One, CORE One L, and CORE One INDX belt alignment and tension
- CORE One L+ belts upgrade
- CORE One L selftest and X-rail settling
- CORE One family X-homing and motor-pulley checks
- GT1.5 21-tooth pulley
- CORE One and CORE One L phase stepping
- Input Shaper
- Firmware 6.8.1 release and CORE One L+ GT1.5 support

Prusa documents both visual and measured extrusion-multiplier methods. This guide recommends the visual/tactile top-surface method for routine filament flow tuning, while using dimensions and fit tests for geometry.

OrcaSlicer

The application comparison in the settings reference is based on OrcaSlicer `v2.4.2`, source commit `8500fcdccaa10b5099ac20d252af3a7c560046f1`. The terminology review used the official OrcaSlicer Wiki at commit `f132cbf1c1e48573d229d1783ee471e61e50aaf3`, dated 2026-09-17. Fixing these revisions makes later UI changes distinguishable from errors in this edition.

- Calibration overview
- Official OrcaSlicer downloads
- Temperature calibration
- Pressure-advance calibration
- Flow ratio and pressure advance settings
- Material information and shrinkage
- Precision and XY compensation
- Tolerance calibration
- Seam settings
- Line-width settings

- Top and bottom shell settings
- Infill and surface pattern reference
- VFA calibration

Firmware Documentation

- Klipper pressure advance
- Klipper resonance compensation
- Klipper accelerometer-based resonance measurement
- Klipper ringing-tower test model
- Marlin Linear Advance

Firmware implementations use different commands and numeric ranges. Never copy a pressure-advance value between firmware families without recalibrating.

Experimental Guides

- Ellis' Print Tuning Guide
- Ellis: extrusion multiplier
- Ellis: determining maximum volumetric flow
- Ellis: common misconceptions
- CNC Kitchen: extrusion-system benchmark tool

These authors emphasize empirical testing and distinguish melt-capacity limits from commanded extrusion. Their exact methods and conclusions should be read at the source before reproducing a formal benchmark.

Dimensional Metrology and Test Artifacts

- **FDM Assessment Protocol — Kickstarter/Autodesk project.** Primary repository and protocol. Used here as the credited source for the tiered X/Y measurement concept and the warning that a meaningful X-versus-Y difference needs investigation.
- **Calistar (formerly Fleur de Cali) — `dirtdigger`.** Primary repository, worksheet, and GPL-3.0 license. Used here as the credited source for the multi-point dimensionality/skew artifact and matched inner/outer measurement approach. Calistar models and worksheets remain the author's work. This repository includes only the separately identified, unmodified GPL-3.0 measurement-layout SVG and its PDF rendering in `assets/third-party/calistar/`; it does not reproduce the Calistar models or worksheet.
- **Teaching Tech Calibration Guide — Teaching Tech.** Primary calibration guide. Used here as the credited source for the distinction between commanded raw motion measured with rigid metrology and dimensions of printed plastic.
- **Califlower Calibration Tool Mk2 — Adam Meadows / Vector 3D.** Official paid product page. Listed as a separately licensed commercial option for dimensional-and-skew calibration. It is not the source of the Calistar measurement diagram and none of its files or instructions are reproduced here.

These sources distinguish characterization of a cooled printed part from calibration of commanded machine motion. Use their models and worksheets under their respective licenses and retain their attribution. Except for the explicitly identified GPL-3.0 figure assets, this repository does not redistribute their files.

Motion-System Mechanics

- CoreXY kinematic theory
- Gates PowerGrip GT3 drive-design manual

- Gates light-power and precision-drive manual
- THK linear-guide mounting and maintenance

The CoreXY transform and belt-drive references support the angle-to-loop and tooth-order method in the motion-diagnostics chapter. They do not establish that a matching frequency identifies one failed component; pulley, idler, belt-plane, rail, and gantry interfaces still require controlled physical inspection and a one-change repeat.

Source Use

No source listed above endorses Tinmans Tuning Guide. This repository does not bundle third-party calibration models or images. Follow each source's license and attribution requirements before redistributing its files.

The settings chapters are an original operational interpretation. Upstream documentation and source code were used to identify controls and behavior, then the material was reorganized around diagnosis, appropriate use, and acceptance criteria. Text was not copied from the OrcaSlicer Wiki.

TinmanX1 Reference

The TinmanX1 comparison and custom-feature documentation are based on source commit `cf0cabb04146c4ff9f21b963517717acfec66055`, whose application compatibility version is `2.4.2`. The installed macOS application examined on 2026-09-21 also reported `CFBundleShortVersionString 2.4.2`. TinmanX1 custom features documented from that revision include continuous-fiber planning, FibreSeek profile contracts, Strength Lens, and experimental Wave Overhangs.

- Official public TinmanX1 downloads

Credits

Tinmans Tuning Guide is an original synthesis built on the work of the open 3D-printing community. It summarizes methods in fresh language and links to the source material instead of reproducing it.

Project Authors

- **William Tinney** initiated the project, supplied the practical tuning questions, performed the physical experiments and measurements, documented print orientation and results, and directed publication.
- **Codex by OpenAI** researched the sources, developed the dependency-aware method, analyzed the experimental evidence, wrote and organized the guide, created the repository, and published this edition.

The worked `20.00 mm -> 20.13 mm -> 100.65% -> 20.00 mm` dimensional example comes from a controlled calibration performed by William Tinney and analyzed with Codex in September 2026.

The motion-diagnostics field photographs and physical FibreSeek and Prusa CORE One L+ experiments were made and supplied by William Tinney. Codex designed the controlled tests, audited the emitted G-code, analyzed the evidence, and authored the worked interpretation. The published photographs have their device and location metadata removed.

Cover Artwork

- The central TinmanX1 thumbnail logo is the original project icon supplied by William Tinney. It is reproduced without redrawing or replacing the mark.
- The circular title composition, calibration-ring motif, typography, layout, and production artwork were designed and prepared by Codex, OpenAI, at William Tinney's direction.
- The editable source, rendered cover, original logo copy, and provenance are retained in the repository's `assets` directory.

Technical Foundations

- **Prusa Research** for its public knowledge base, especially its extrusion multiplier, drying, warping, first-layer, and material guidance.
- **SoftFever and the OrcaSlicer contributors** for the integrated calibration tools and documentation covering temperature, volumetric speed, flow ratio, pressure advance, tolerance, shrinkage, precision, seams, and surfaces.
- **TinmanX1 contributors** for the source implementation used to document its continuous-fiber planning, FibreSeek contracts, Strength Lens, and Wave Overhang extensions. William Tinney directed the practical scope of this documentation; Codex authored the plain-language operational reference.
- **Kevin O'Connor and the Klipper contributors** for the pressure-advance implementation and calibration documentation.
- **The Marlin contributors** for Linear Advance and its calibration documentation.
- **Andrew Ellis** for Ellis' Print Tuning Guide, especially the practical visual extrusion-multiplier method, dimensional-calibration cautions, and maximum-volumetric-flow guidance.
- **Stefan Hermann of CNC Kitchen** for empirical extrusion-system and hotend flow benchmarking. His work is part of the test lineage cited by both Ellis' guide and the broader calibration community.
- **The PrusaSlicer and SuperSlicer contributors** for the slicer features and calibration lineage inherited by later open-source slicers.

Third-Party Figure Assets

- **Calistar (formerly Fleur de Cali) by dirtdigger** for the GPL-3.0 measurement-layout SVG reproduced as a raster figure in the dimensional accuracy chapter. The original SVG and its GPL-3.0 notice are retained in `assets/third-party/calistar/`; those files are not relicensed under this guide's CC BY 4.0 license.

- **Adam Meadows / Vector 3D** for the official Califlower Calibration Tool Mk2 reference. Califlower is a separately licensed commercial product; this guide names and links it as an alternative but reproduces none of its files or instructions.

Specific links are collected in SOURCES.md.

Attribution Policy

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Contributors should identify the source of any new method, formula, diagram, or test model in the same pull request that introduces it.

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